

Signal and Systems

ECECC05

Syllabus

Unit-I:

- Concept definition & classification of signals– Continuous time (CT) and Discrete Time (DT) signals, Standard signals- Step, Ramp, Pulse, Impulse, Real and complex exponentials and Sinusoids, Transformation of independent variables, Classification of systems and their properties

Unit-II:

- Convolution and its properties in continuous and discrete domain, LTI Systems and its properties, impulse response, step response, Linear convolution in continuous and discrete domain using graphical method, using general formula and matrix method.

Unit-III:

- Need and applications of Fourier series, Fourier series representation of continuous time and discrete time signals using trigonometric method and exponential method and its properties.
- Continuous time Fourier Transform and its properties, Discrete time Fourier Transform and its properties. Magnitude and phase representations of frequency response of LTI systems. Analysis and characterization of LTI systems using Differential Equations and Difference equations.

Unit-IV:

- Laplace transform and its properties, concept of ROC and its properties. Inverse-Laplace transform. Unilateral Laplace transform and its applications. Computation of impulse response, total response (zero state and zero input response) & transfer function using Laplace transform. Stability and causality of continuous-time LTI Systems.

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Unit-V:

- Z-Transform and its properties, ROC and its properties, Inverse Z-transform; Power series expansion and partial fraction expansion. Relationship between Z-transform, Fourier transform and Laplace transform. Stability and causality of discrete-time LTI System. Unilateral Z transform and its applications. Computation of impulse response, total response (Zero state and Zero input response) & Transfer function using Z-Transform.

Text Books:

- [T1] Alan V. Oppenheim, Alan S. Willsky, S. Hamid Nawab, Signals and Systems Prentice Hall India, 2nd Edition, 2009.

Reference Books:

- [R1] Simon Haykin and Barry Van Veen, “Signals and Systems”, John Willey & Sons, Inc, second edition 2013.
- [R2] Tarun Kumar Rawat, Signals and Systems, Oxford University Press, 1st Edition, 2010.

Introduction

- Signals are represented mathematically as functions of one or more independent variables.
- For example, a speech signal can be represented mathematically by acoustic pressure as a function of time, and a picture can be represented by brightness as a function of two spatial variables
- We focus our attention on signals involving a single independent variable, we will generally refer to the independent variable as time.
- Throughout this subject we will be considering two basic types of signals: continuous time signals (t) and discrete-time signals $[n]$.
- In the case of continuous-time signals the independent variable is continuous
- Discrete-time signals are defined only at discrete times

Notation— $x(t)$ —Continuous time, $x[n]$ —Discrete time

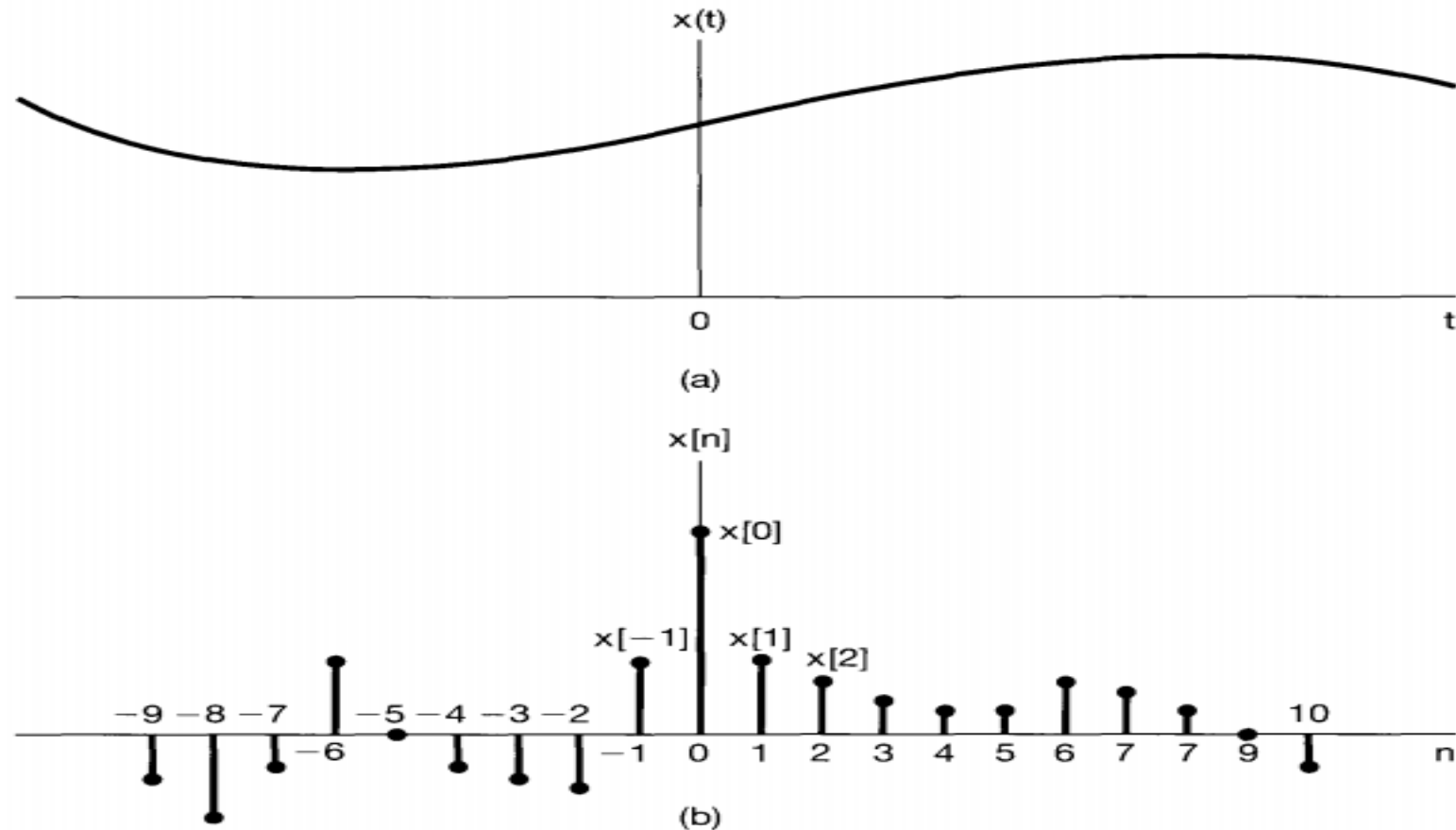


Figure 1.7 Graphical representations of (a) continuous-time and (b) discrete-time signals.

Energy Signal and Power Signal

- Signals with Finite energy are known as Energy signals

$$E_{\infty} \triangleq \lim_{T \rightarrow \infty} \int_{-T}^T |x(t)|^2 dt = \int_{-\infty}^{+\infty} |x(t)|^2 dt, \quad E_{\infty} \triangleq \lim_{N \rightarrow \infty} \sum_{n=-N}^{+N} |x[n]|^2 = \sum_{n=-\infty}^{+\infty} |x[n]|^2.$$

- Signals with Finite power are known as Power signals

$$P_{\infty} \triangleq \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T |x(t)|^2 dt$$

$$P_{\infty} \triangleq \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^{+N} |x[n]|^2$$

- Power of an energy signal is zero
Energy of power signal is infinite.

$$P_{\infty} = \lim_{T \rightarrow \infty} \frac{E_{\infty}}{2T} = 0.$$

There are also signals for which neither P_x nor E_x are finite. A simple example is the signal $x(t) = t$.

Analog and Digital Signals

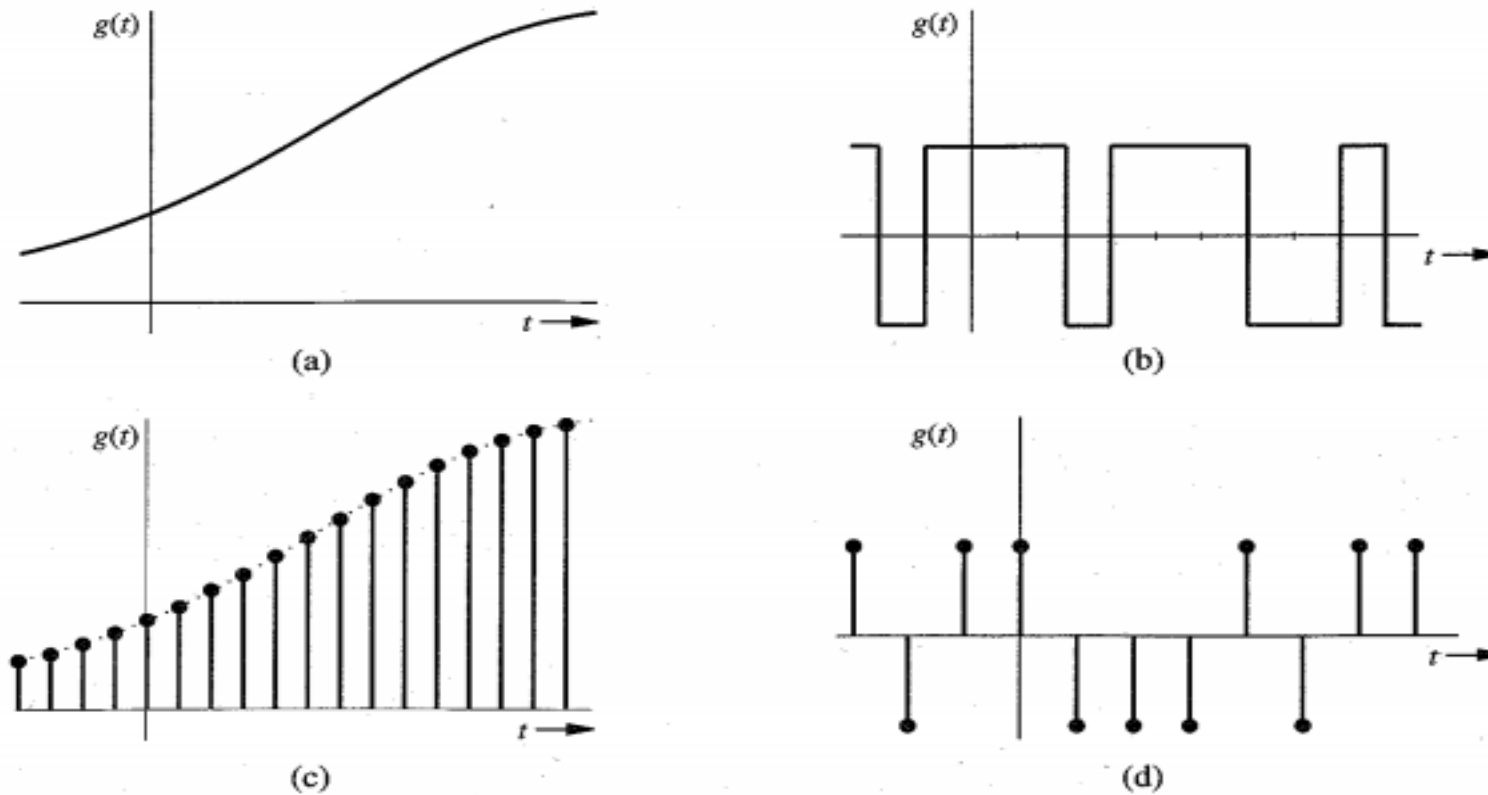


Figure 2.5 Examples of signals. (a) Analog, continuous time. (b) Digital, continuous time. (c) Analog, discrete time. (d) Digital, discrete time.

Periodic Signals and Aperiodic Signals

A signal $g(t)$ is said to be **periodic** if for some positive constant T_0 ,

$$g(t) = g(t + T_0) \quad \text{for all } t \quad (2.7)$$

The *smallest* value of T_0 that satisfies the periodicity condition (2.7) is the **period** of $g(t)$. The signal in Fig. 2.3b is a periodic signal with period 2. A signal is **aperiodic** if it is not periodic. The signal in Fig. 2.3a is aperiodic.

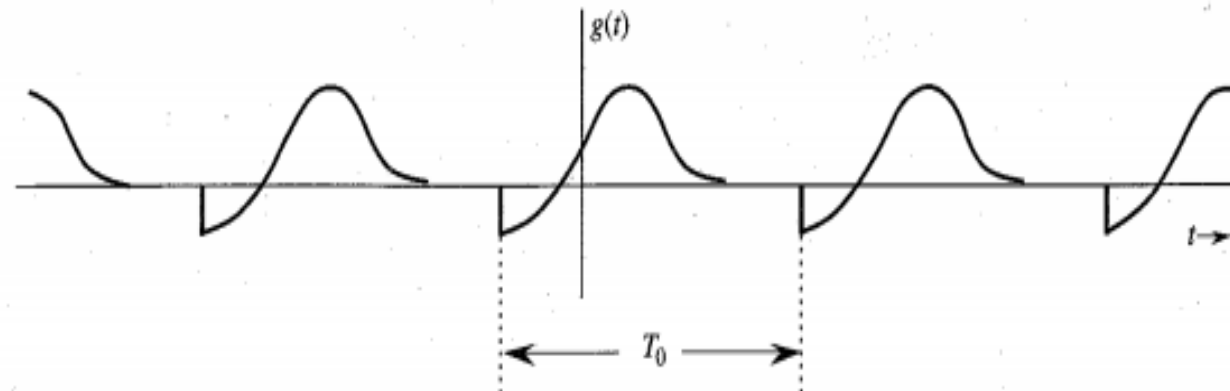


Figure 2.6 Periodic signal of period T_0 .

Deterministic and Random Signals

A signal whose physical description is known completely, in either a mathematical form or a graphical form, is a **deterministic signal**. If a signal is known only in terms of probabilistic description, such as mean value, mean squared value, and so on, rather than its complete mathematical or graphical description, is a **random signal**. Most of the noise signals encountered in practice are random signals. All message signals are random signals because, as will be shown later, a signal, to convey information, must have some uncertainty (randomness) about

Continuous-Time Complex Exponential and Sinusoidal Signals

$$x(t) = Ce^{at},$$

where C and a are, in general, complex numbers. Depending upon the values of these parameters, the complex exponential can exhibit several different characteristics.

Real Exponential Signals
(C and a both are real)



Figure 1.19 Continuous-time real exponential $x(t) = Ce^{at}$: (a) $a > 0$; (b) $a < 0$.

Periodic Complex Exponential and Sinusoidal Signals (a to be purely imaginary)

$$x(t) = e^{j\omega_0 t}.$$

An important property of this signal is that it is periodic. To verify this, we recall from eq. (1.11) that $x(t)$ will be periodic with period T if

$$e^{j\omega_0 t} = e^{j\omega_0(t+T)}. \quad (1.22)$$

Or, since

$$e^{j\omega_0(t+T)} = e^{j\omega_0 t} e^{j\omega_0 T},$$

it follows that for periodicity, we must have

$$e^{j\omega_0 T} = 1. \quad (1.23)$$

If $\omega_0 = 0$, then $x(t) = 1$, which is periodic for any value of T . If $\omega_0 \neq 0$, then the fundamental period T_0 of $x(t)$ —that is, the smallest positive value of T for which eq. (1.23) holds—is

$$T_0 = \frac{2\pi}{|\omega_0|}. \quad (1.24)$$

Thus, the signals $e^{j\omega_0 t}$ and $e^{-j\omega_0 t}$ have the same fundamental period.

A signal closely related to the periodic complex exponential is the *sinusoidal signal*

$$x(t) = A \cos(\omega_0 t + \phi), \quad (1.25)$$

General Complex Exponential Signals

$$C = |C|e^{j\theta}$$

and

$$a = r + j\omega_0.$$

Then

$$Ce^{at} = |C|e^{j\theta}e^{(r+j\omega_0)t} = |C|e^{rt}e^{j(\omega_0 t + \theta)}. \quad (1.42)$$

Using Euler's relation, we can expand this further as

$$Ce^{at} = |C|e^{rt} \cos(\omega_0 t + \theta) + j|C|e^{rt} \sin(\omega_0 t + \theta). \quad (1.43)$$

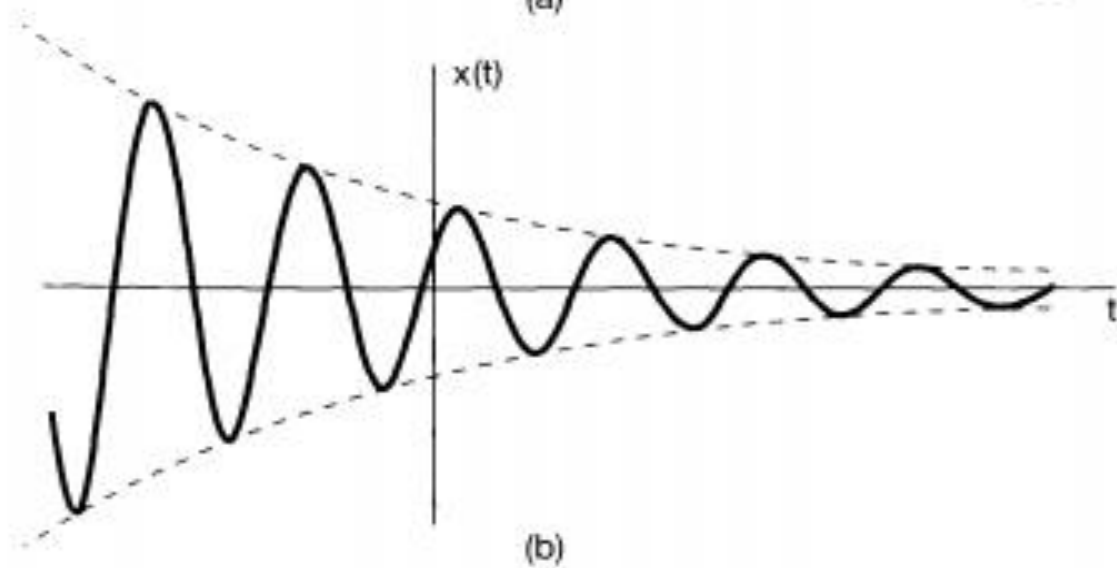
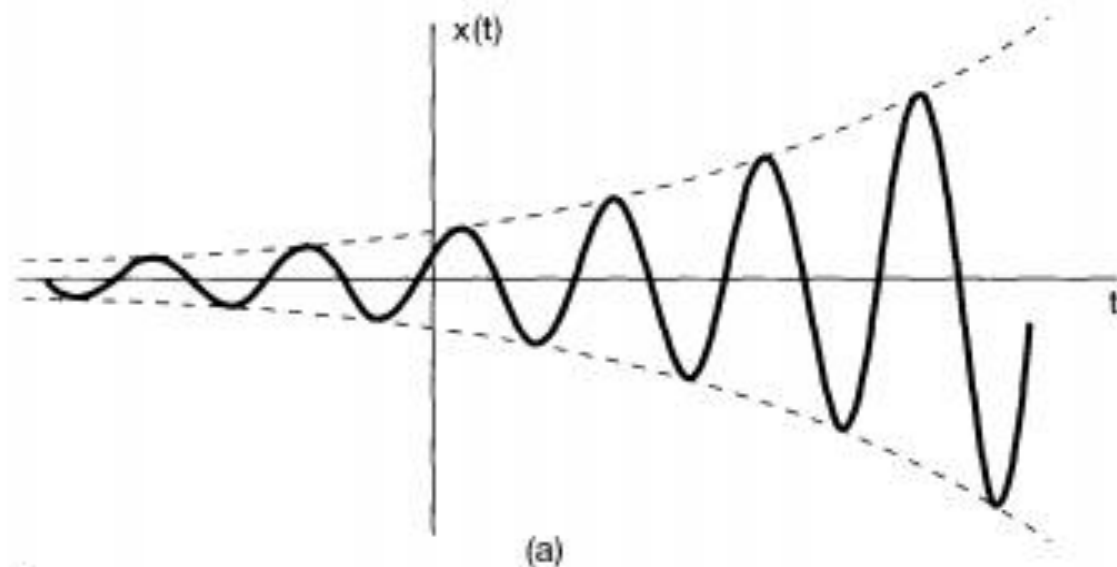


Figure 1.23 (a) Growing sinusoidal signal $x(t) = Ce^{rt} \cos(\omega_0 t + \theta)$, $r > 0$; (b) decaying sinusoid $x(t) = Ce^{rt} \cos(\omega_0 t + \theta)$, $r < 0$.

Properties:

- **Systems with and without Memory**

A system is said to be *memoryless* if its output for each value of the independent variable at a given time is dependent only on the input at that same time. For example, the system specified by the relationship

$$y[n] = (2x[n] - x^2[n])^2$$

- A system with memory is

$$y[n] = \sum_{k=-\infty}^n x[k],$$

- **Invertibility and Inverse Systems**

A system is said to be *invertible* if distinct inputs lead to distinct outputs.

An example of an invertible continuous-time system is

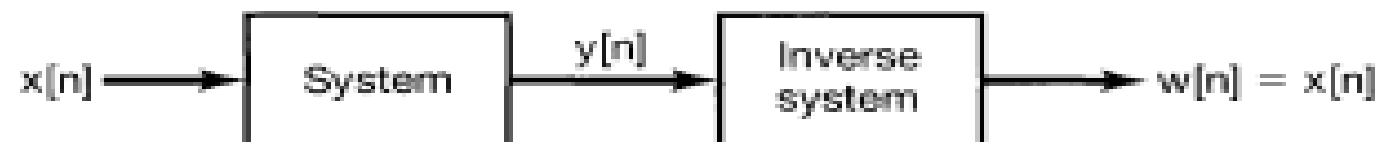
$$y(t) = 2x(t), \quad (1.97)$$

for which the inverse system is

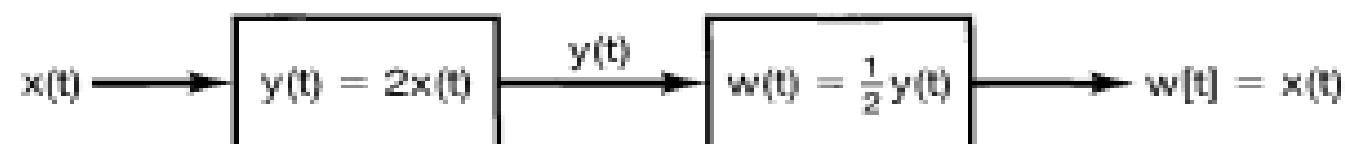
$$w(t) = \frac{1}{2}y(t). \quad (1.98)$$

- Not invertible system

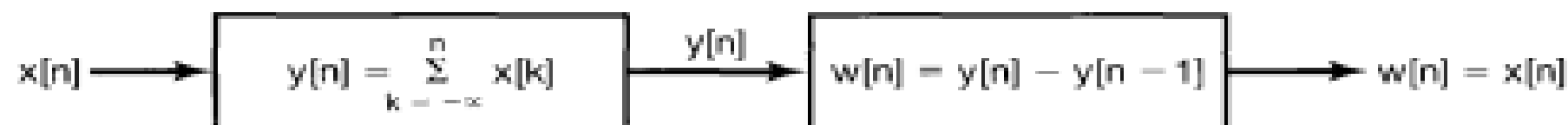
$$y(t) = x^2(t),$$



(a)



(b)



(c)

Figure 1.45 Concept of an inverse system for: (a) a general invertible system; (b) the invertible system described by eq. (1.97); (c) the invertible system defined in eq. (1.92).

- **Causality**

A system is *causal* if the output at any time depends only on values of the input at the present time and in the past. Such a system is often referred to as being *nonanticipative*

$$y[n] = \sum_{k=-\infty}^n x[k], \quad (1.92)$$

and a second example is a *delay*

$$y[n] = x[n - 1]. \quad (1.93)$$

- $Y[n]=x[n+1]$

LTI Systems

- **The Representation of Discrete-Time Signals in Terms of Impulses:**

$$x[-1]\delta[n+1] = \begin{cases} x[-1], & n = -1 \\ 0, & n \neq -1 \end{cases},$$

$$x[0]\delta[n] = \begin{cases} x[0], & n = 0 \\ 0, & n \neq 0 \end{cases},$$

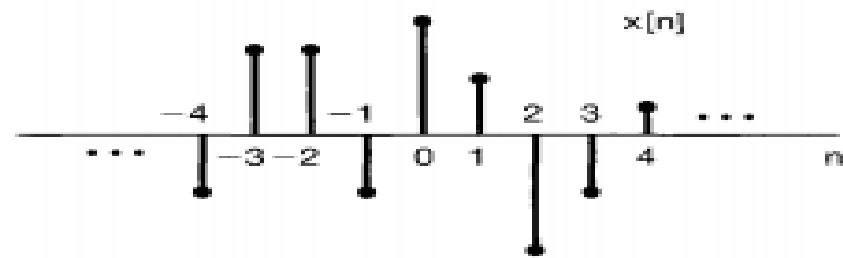
$$x[1]\delta[n-1] = \begin{cases} x[1], & n = 1 \\ 0, & n \neq 1 \end{cases}.$$

- **In general, we can write**

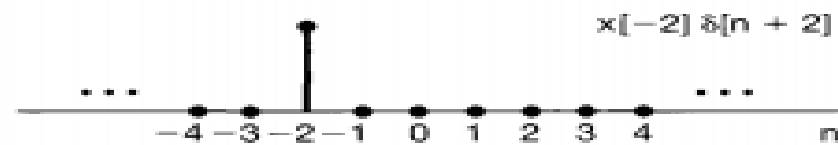
$$\begin{aligned} x[n] = & \dots + x[-3]\delta[n+3] + x[-2]\delta[n+2] + x[-1]\delta[n+1] + x[0]\delta[n] \\ & + x[1]\delta[n-1] + x[2]\delta[n-2] + x[3]\delta[n-3] + \dots \end{aligned}$$

• In compact form:
$$x[n] = \sum_{k=-\infty}^{+\infty} x[k] \delta[n - k].$$

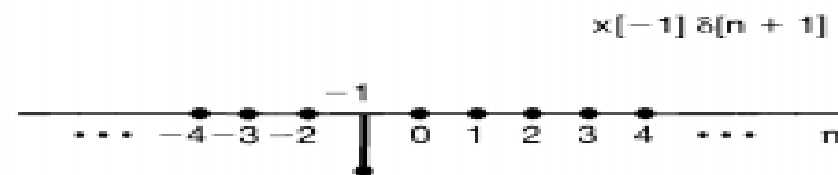
Representation of arbitrary sequence in terms of weighted shifted impulses.



(a)



(b)



(c)

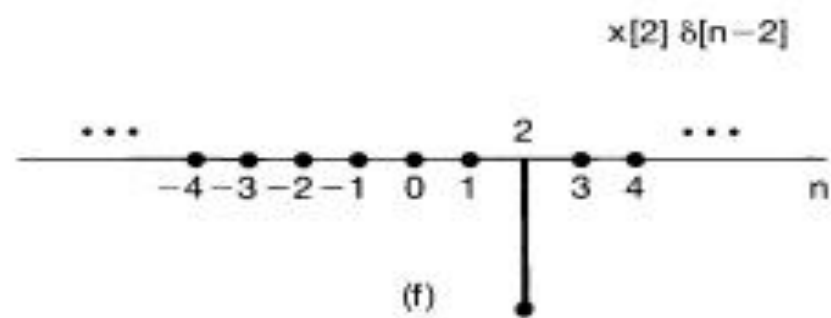
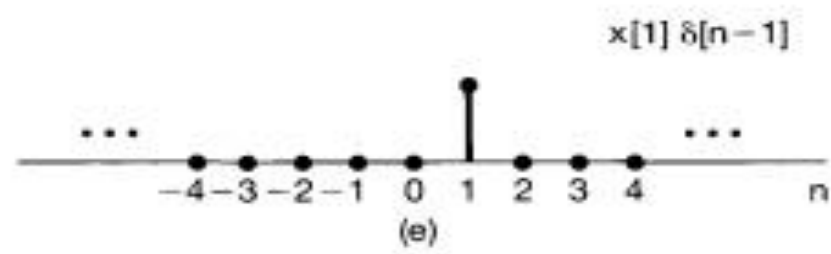
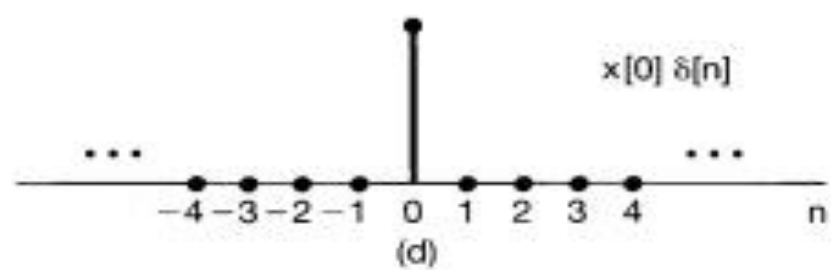


Figure 2.1 Decomposition of a discrete-time signal into a weighted sum of shifted impulses.

- **The Discrete-Time Unit Impulse Response and the Convolution Sum Representation of LTI Systems:**

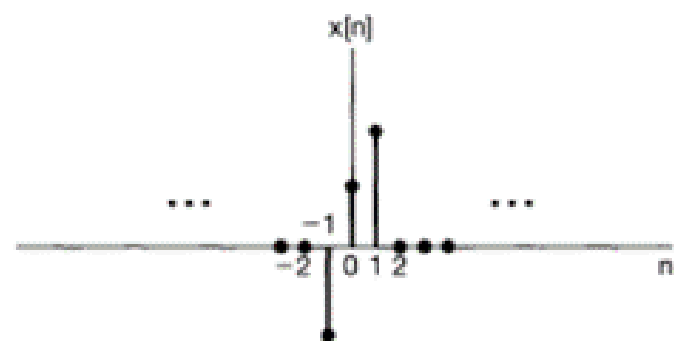
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More specifically, consider the response of a linear (but possibly time-varying) system to an arbitrary input $x[n]$. We can represent the input through eq. (2.2) as a linear combination of shifted unit impulses. Let $h_k[n]$ denote the response of the linear system to the shifted unit impulse $\delta[n - k]$. Then, from the superposition property for a linear system [eqs. (1.123) and (1.124)], the response $y[n]$ of the linear system to the input $x[n]$ in eq. (2.2) is simply the weighted linear combination of these basic responses. That is, with the input $x[n]$ to a linear system expressed in the form of eq. (2.2), the output $y[n]$ can be expressed as

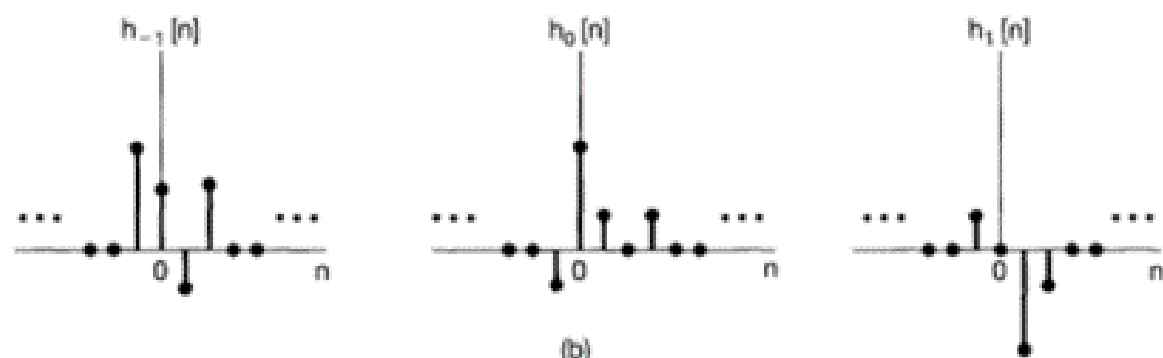
$$y[n] = \sum_{k=-\infty}^{+\infty} x[k]h_k[n]. \quad (2.3)$$

- If we know the responses of shifted unit impulses, we can get the output of system for any arbitrary input.
- Ex: Let $x[n]$ applied to a system whose response is known for

$\delta[n + 1]$, $\delta[n]$, and $\delta[n - 1]$, responses $h_{-1}[n]$, $h_0[n]$, and $h_1[n]$!



(a)



(b)

- For time invariant system,

$$h_k[n] = h_0[n - k].$$

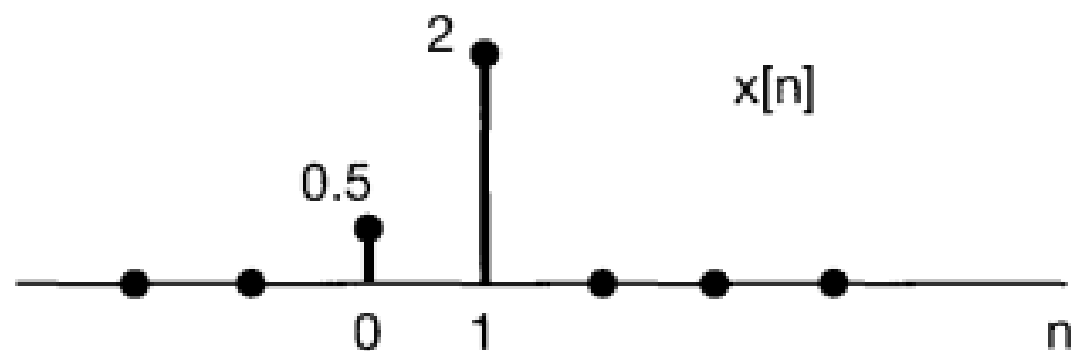
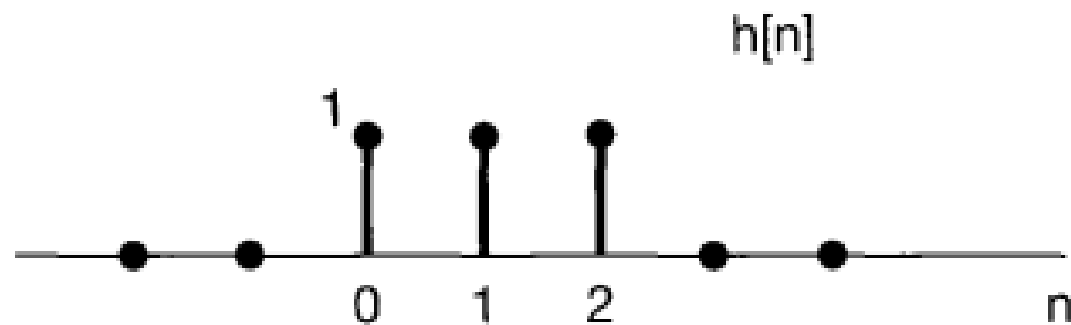
- For LTI system, we can write the output of a system
-

$$y[n] = \sum_{k=-\infty}^{+\infty} x[k]h[n - k].$$

This result is referred to as the *convolution sum* or *superposition sum*,

$$y[n] = x[n] * h[n].$$

Example: Find $y[n]$ for given $x[n]$ and $h[n]$

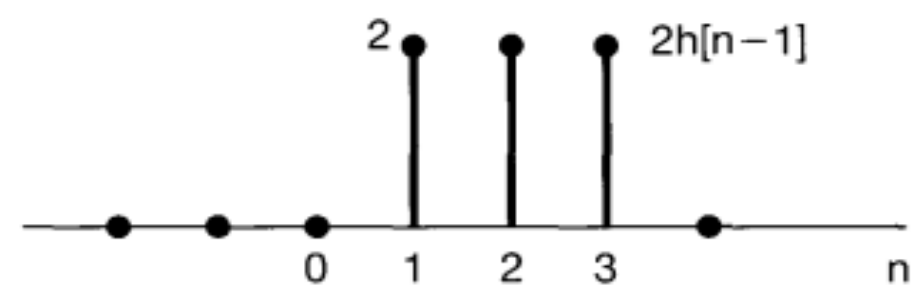
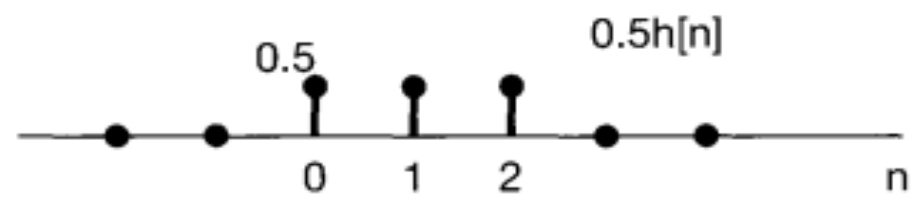


(a)

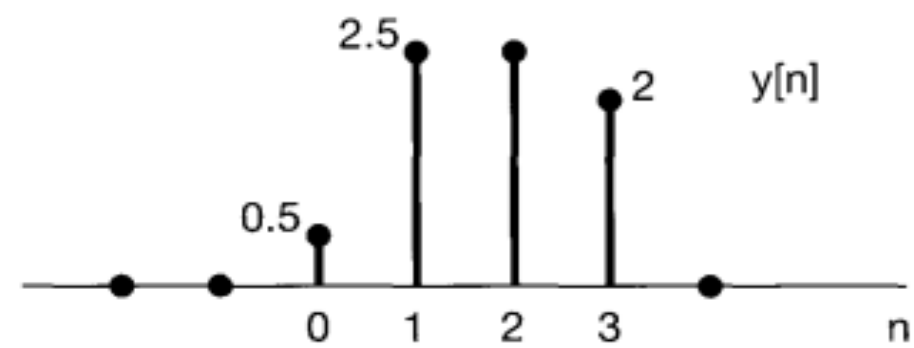
Solution

$$y[n] = x[n] * h[n].$$

$$y[n] = \sum_{k=-\infty}^{+\infty} x[k]h[n-k].$$



(b)



(c)

Linear Convolution using Matrix Method

When the sequences $h[n]$ and $x[n]$ are represented as matrices, the linear convolution operation can be equivalently represented as

$$y = h * x = x * h = \text{Toeplitz}(h) X$$

Let us take $h[n] = [h_0, h_1, h_2, h_3]$

and $x[n] = [x_0, x_1, x_2]$

The convolution $h[n] * x[n]$ is given by

- The convolution $h[n]*x[n]$ is given by

$$y[n]=h[n]*x[n]$$

$$y[n] = \sum_{k=-\infty}^{+\infty} x[k]h[n-k].$$

Equivalent representation of the above convolution can be written as

$$X[n]=x[0], x[1], x[2]; h[n]= h[0], h[1], h[2], h[3]$$

$$\begin{pmatrix} y[0] \\ y[1] \\ y[2] \\ y[3] \\ y[4] \\ y[5] \end{pmatrix} = \begin{pmatrix} h[0] & 0 & 0 \\ h[1] & h[0] & 0 \\ h[2] & h[1] & h[0] \\ h[3] & h[2] & h[1] \\ 0 & h[3] & h[2] \\ 0 & 0 & h[3] \end{pmatrix} \begin{pmatrix} x[0] \\ x[1] \\ x[2] \end{pmatrix}$$

Toeplitz Matrix

$$A = \begin{bmatrix} a_0 & a_{-1} & a_{-2} & \cdots & \cdots & a_{-(n-1)} \\ a_1 & a_0 & a_{-1} & \ddots & & \vdots \\ a_2 & a_1 & \ddots & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & a_{-1} & a_{-2} \\ \vdots & & \ddots & a_1 & a_0 & a_{-1} \\ a_{n-1} & \cdots & \cdots & a_2 & a_1 & a_0 \end{bmatrix}$$

In general

$$y = h * x = \begin{bmatrix} h_1 & 0 & \cdots & 0 & 0 \\ h_2 & h_1 & & \vdots & \vdots \\ h_3 & h_2 & \cdots & 0 & 0 \\ \vdots & h_3 & \cdots & h_1 & 0 \\ h_{m-1} & \vdots & \ddots & h_2 & h_1 \\ h_m & h_{m-1} & & \vdots & h_2 \\ 0 & h_m & \ddots & h_{m-2} & \vdots \\ 0 & 0 & \cdots & h_{m-1} & h_{m-2} \\ \vdots & \vdots & & h_m & h_{m-1} \\ 0 & 0 & 0 & \cdots & h_m \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix}$$

Example:

- $X[n] = [1 \ 2 \ 3 \ 4]$; $h[n] = [7 \ 6 \ 5]$ calculate $y[n]$

$$\begin{pmatrix} y[0] \\ y[1] \\ y[2] \\ y[3] \\ y[4] \\ y[5] \end{pmatrix} = \begin{pmatrix} 7 & 0 & 0 & 0 \\ 6 & 7 & 0 & 0 \\ 5 & 6 & 7 & 0 \\ 0 & 5 & 6 & 7 \\ 0 & 0 & 5 & 6 \\ 0 & 0 & 0 & 5 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix} = \begin{pmatrix} 7 \\ 20 \\ 38 \\ 56 \\ 39 \\ 20 \end{pmatrix}$$

Eigen Functions

- We represent the signal as a linear combination of some basic function like impulse function.....Convolution
- Alternative representation is done by using complex exponentials.
- **THE RESPONSE OF LTI SYSTEMS TO COMPLEX EXPONENTIALS**
The set of basic signals can be used to construct a broad and useful class of signals.
- The response of an LTI system to each signal should be simple enough in structure to provide us with a convenient representation for the response of the system to any signal constructed as a linear combination of the basic signals.

The importance of complex exponentials in the study of LTI systems stems from the fact that the response of an LTI system to a complex exponential input is the same complex exponential with only a change in amplitude; that is,

$$\text{continuous time: } e^{st} \longrightarrow H(s)e^{st}, \quad (3.1)$$

$$\text{discrete time: } z^n \longrightarrow H(z)z^n, \quad (3.2)$$

where the complex amplitude factor $H(s)$ or $H(z)$ will in general be a function of the complex variable s or z . A signal for which the system output is a (possibly complex)

constant times the input is referred to as an *eigenfunction* of the system, and the amplitude factor is referred to as the system's *eigenvalue*.

To show that complex exponentials are indeed eigenfunctions of LTI systems, let us consider a continuous-time LTI system with impulse response $h(t)$. For an input $x(t)$, we can determine the output through the use of the convolution integral, so that with $x(t) = e^{st}$

$$\begin{aligned} y(t) &= \int_{-\infty}^{+\infty} h(\tau)x(t - \tau) d\tau \\ &= \int_{-\infty}^{+\infty} h(\tau)e^{s(t-\tau)} d\tau. \end{aligned} \tag{3.3}$$

Expressing $e^{s(t-\tau)}$ as $e^{st}e^{-s\tau}$, and noting that e^{st} can be moved outside the integral, we see that eq. (3.3) becomes

$$y(t) = e^{st} \int_{-\infty}^{+\infty} h(\tau)e^{-s\tau} d\tau. \tag{3.4}$$

Assuming that the integral on the right-hand side of eq. (3.4) converges, the response to e^{st} is of the form

$$y(t) = H(s)e^{st}, \quad (3.5)$$

where $H(s)$ is a complex constant whose value depends on s and which is related to the system impulse response by

$$H(s) = \int_{-\infty}^{+\infty} h(\tau)e^{-s\tau} d\tau. \quad (3.6)$$

Hence, we have shown that complex exponentials are eigenfunctions of LTI systems. The constant $H(s)$ for a specific value of s is then the eigenvalue associated with the eigenfunction e^{st} .

Hence, we have shown that complex exponentials are eigenfunctions of LTI systems. The constant $H(s)$ for a specific value of s is then the eigenvalue associated with the eigenfunction e^{st} .

In an exactly parallel manner, we can show that complex exponential sequences are eigenfunctions of discrete-time LTI systems. That is, suppose that an LTI system with impulse response $h[n]$ has as its input the sequence

$$x[n] = z^n, \quad (3.7)$$

where z is a complex number. Then the output of the system can be determined from the convolution sum as

$$\begin{aligned} y[n] &= \sum_{k=-\infty}^{+\infty} h[k]x[n-k] \\ &= \sum_{k=-\infty}^{+\infty} h[k]z^{n-k} = z^n \sum_{k=-\infty}^{+\infty} h[k]z^{-k}. \end{aligned} \quad (3.8)$$

From this expression, we see that if the input $x[n]$ is the complex exponential given by eq. (3.7), then, assuming that the summation on the right-hand side of eq. (3.8) converges, the output is the same complex exponential multiplied by a constant that depends on the value of z . That is,

$$y[n] = H(z)z^n, \quad (3.9)$$

where

$$H(z) = \sum_{k=-\infty}^{+\infty} h[k]z^{-k}. \quad (3.10)$$

Consequently, as in the continuous-time case, complex exponentials are eigenfunctions of discrete-time LTI systems. The constant $H(z)$ for a specified value of z is the eigenvalue associated with the eigenfunction z^n .

For the analysis of LTI systems, the usefulness of decomposing more general signals in terms of eigenfunctions can be seen from an example. Let $x(t)$ correspond to a linear combination of three complex exponentials; that is,

$$x(t) = a_1 e^{s_1 t} + a_2 e^{s_2 t} + a_3 e^{s_3 t}. \quad (3.11)$$

From the eigenfunction property, the response to each separately is

$$\begin{aligned} a_1 e^{s_1 t} &\longrightarrow a_1 H(s_1) e^{s_1 t}, \\ a_2 e^{s_2 t} &\longrightarrow a_2 H(s_2) e^{s_2 t}, \\ a_3 e^{s_3 t} &\longrightarrow a_3 H(s_3) e^{s_3 t}, \end{aligned}$$

and from the superposition property the response to the sum is the sum of the responses, so that

$$y(t) = a_1 H(s_1) e^{s_1 t} + a_2 H(s_2) e^{s_2 t} + a_3 H(s_3) e^{s_3 t}. \quad (3.12)$$

More generally, in continuous time, eq. (3.5), together with the superposition property, implies that the representation of signals as a linear combination of complex exponentials leads to a convenient expression for the response of an LTI system. Specifically, if the input to a continuous-time LTI system is represented as a linear combination of complex exponentials, that is, if

$$x(t) = \sum_k a_k e^{s_k t}, \quad (3.13)$$

then the output will be

$$y(t) = \sum_k a_k H(s_k) e^{s_k t}. \quad (3.14)$$

In an exactly analogous manner, if the input to a discrete-time LTI system is represented as a linear combination of complex exponentials, that is, if

$$x[n] = \sum_k a_k z_k^n, \quad (3.15)$$

then the output will be

$$y[n] = \sum_k a_k H(z_k) z_k^n. \quad (3.16)$$

Example 3.1

As an illustration of eqs. (3.5) and (3.6), consider an LTI system for which the input $x(t)$ and output $y(t)$ are related by a time shift of 3, i.e.,

$$y(t) = x(t - 3). \quad (3.17)$$

If the input to this system is the complex exponential signal $x(t) = e^{j2t}$, then, from eq. (3.17),

$$y(t) = e^{j2(t-3)} = e^{-j6} e^{j2t}. \quad (3.18)$$

Equation (3.18) is in the form of eq. (3.5), as we would expect, since e^{j2t} is an eigenfunction. The associated eigenvalue is $H(j2) = e^{-j6}$. It is straightforward to confirm eq. (3.6) for this example. Specifically, from eq. (3.17), the impulse response of the system is $h(t) = \delta(t - 3)$. Substituting into eq. (3.6), we obtain

$$H(s) = \int_{-\infty}^{+\infty} \delta(\tau - 3) e^{-s\tau} d\tau = e^{-3s},$$

so that $H(j2) = e^{-j6}$.

FOURIER SERIES REPRESENTATION OF CONTINUOUS-TIME PERIODIC SIGNALS

- A signal is periodic if, for some positive value of T , $x(t) = x(t + T)$ for all t .

The fundamental period of $x(t)$ is the minimum positive, nonzero value of T for which the given relation is satisfied, and the value

$\omega_0 = 2\pi/T$ is referred to as the fundamental frequency

Basic periodic signals are:

$$x(t) = \cos \omega_0 t$$

and the periodic complex exponential

$$x(t) = e^{j\omega_0 t}.$$

- The set of harmonically related complex exponentials are

$$\phi_k(t) = e^{jk\omega_0 t} = e^{jk(2\pi/T)t}, \quad k = 0, \pm 1, \pm 2, \dots$$

- A linear combination of harmonically related exponentials is also periodic with period T.

$$x(t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t} = \sum_{k=-\infty}^{+\infty} a_k e^{jk(2\pi/T)t}$$

- This representation of x(t) is known as Fourier series representation.

Consider a periodic signal $x(t)$, with fundamental frequency 2π , that is expressed in the form of eq. (3.25) as

$$x(t) = \sum_{k=-3}^{+3} a_k e^{jk2\pi t}, \quad (3.26)$$

where

$$a_0 = 1,$$

$$a_1 = a_{-1} = \frac{1}{4},$$

$$a_2 = a_{-2} = \frac{1}{2},$$

$$a_3 = a_{-3} = \frac{1}{3}.$$

$$\begin{aligned}x(t) = & 1 + \frac{1}{4}(e^{j2\pi t} + e^{-j2\pi t}) + \frac{1}{2}(e^{j4\pi t} + e^{-j4\pi t}) \\& + \frac{1}{3}(e^{j6\pi t} + e^{-j6\pi t}).\end{aligned}$$

Equivalently, using Euler's relation, we can write $x(t)$ in the form

$$x(t) = 1 + \frac{1}{2} \cos 2\pi t + \cos 4\pi t + \frac{2}{3} \cos 6\pi t.$$

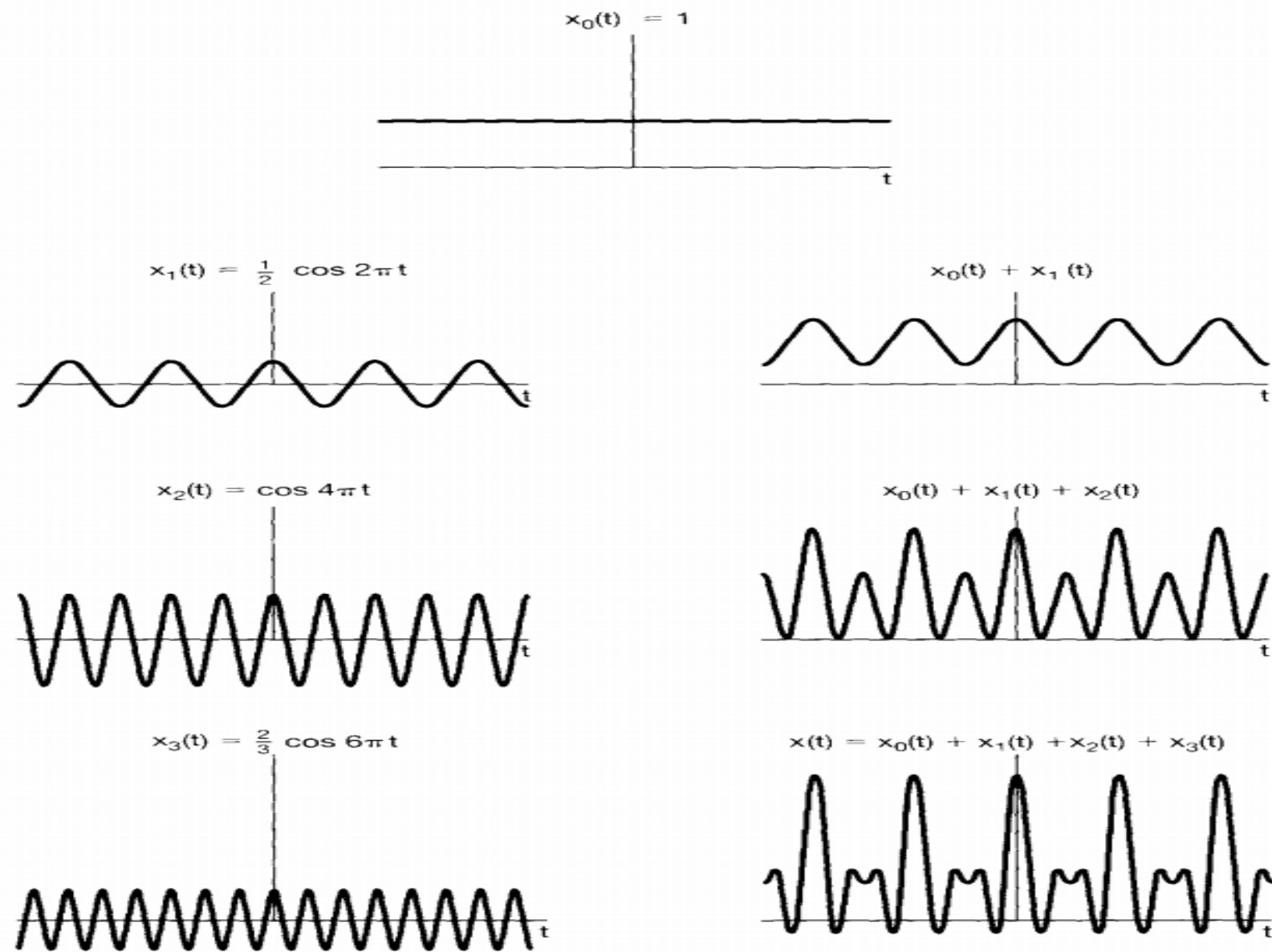


Figure 3.4 Construction of the signal $x(t)$ in Example 3.2 as a linear combination of harmonically related sinusoidal signals.

Real periodic signals in continuous time

$$x(t) = a_0 + 2 \sum_{k=1}^{\infty} A_k \cos(k\omega_0 t + \theta_k).$$

$$x(t) = a_0 + 2 \sum_{k=1}^{\infty} [B_k \cos k\omega_0 t - C_k \sin k\omega_0 t].$$

Determination of the Fourier Series Representation of a Periodic Signal

$$\begin{aligned}x[0] &= \sum_{k=\langle N \rangle} a_k, \\x[1] &= \sum_{k=\langle N \rangle} a_k e^{j2\pi k/N}, \\&\vdots \\x[N-1] &= \sum_{k=\langle N \rangle} a_k e^{j2\pi k(N-1)/N}.\end{aligned}$$

The basis for this result is the fact, shown in Problem 3.54, that

$$\sum_{n=\langle N \rangle} e^{jk(2\pi/N)n} = \begin{cases} N, & k = 0, \pm N, \pm 2N, \dots \\ 0, & \text{otherwise} \end{cases}.$$

Now consider the Fourier series representation of eq. (3.88). Multiplying both sides by $e^{-jr(2\pi/N)n}$ and summing over N terms, we obtain

$$\sum_{n=\langle N \rangle} x[n]e^{-jr(2\pi/N)n} = \sum_{n=\langle N \rangle} \sum_{k=\langle N \rangle} a_k e^{j(k-r)(2\pi/N)n}. \quad (3.91)$$

Interchanging the order of summation on the right-hand side, we have

$$\sum_{n=\langle N \rangle} x[n]e^{-jr(2\pi/N)n} = \sum_{k=\langle N \rangle} a_k \sum_{n=\langle N \rangle} e^{j(k-r)(2\pi/N)n}.$$

- The innermost sum on the right-hand side of eq. (3.92) equals N if $k = r$ and 0 if $k \neq r$. The right-hand side of eq. (3.92) then reduces to Na_r and we have

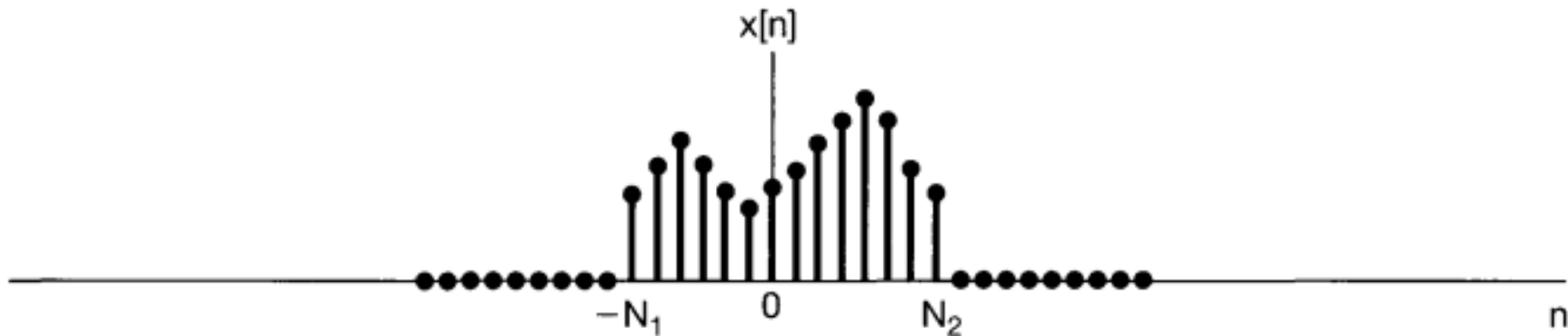
$$a_r = \frac{1}{N} \sum_{n=\langle N \rangle} x[n] e^{-jr(2\pi/N)n}.$$

$$\begin{aligned} x[n] &= \sum_{k=\langle N \rangle} a_k e^{jk\omega_0 n} = \sum_{k=\langle N \rangle} a_k e^{jk(2\pi/N)n}, \\ a_k &= \frac{1}{N} \sum_{n=\langle N \rangle} x[n] e^{-jk\omega_0 n} = \frac{1}{N} \sum_{n=\langle N \rangle} x[n] e^{-jk(2\pi/N)n}. \end{aligned}$$

Discrete Time Fourier Transform

Let us now examine the Fourier series representation of $\tilde{x}[n]$. Specifically, from eqs. (3.94) and (3.95), we have

$$\tilde{x}[n] = \sum_{k=\langle N \rangle} a_k e^{jk(2\pi/N)n}, \quad (5.1)$$



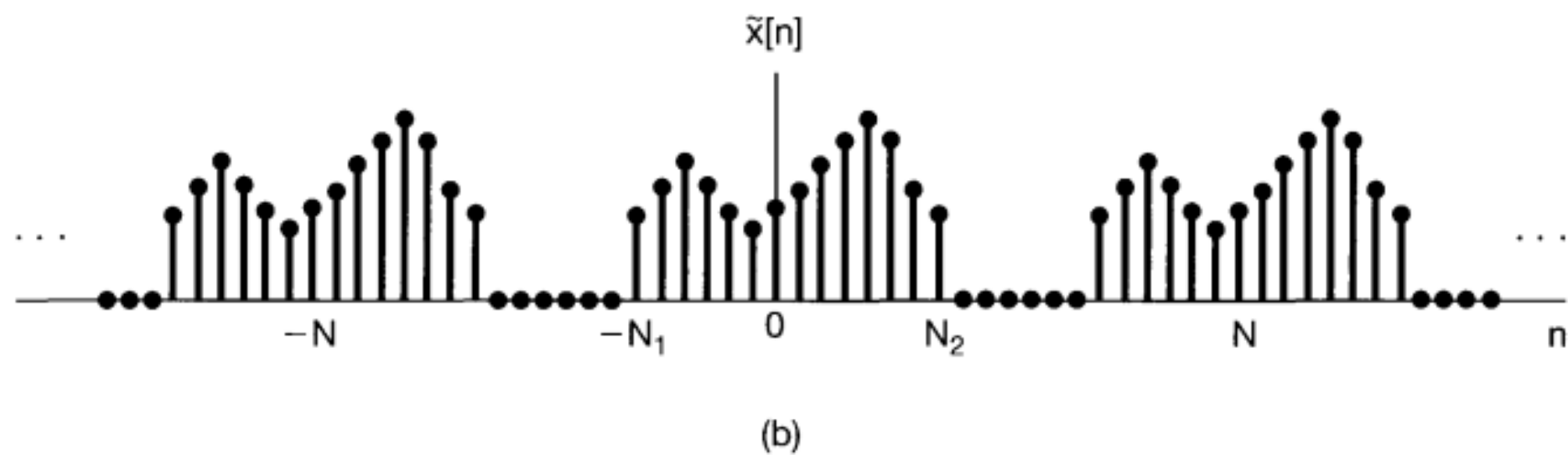


Figure 5.1 (a) Finite-duration signal $x[n]$; (b) periodic signal $\tilde{x}[n]$ constructed to be equal to $x[n]$ over one period.

$$a_k = \frac{1}{N} \sum_{n=\langle N \rangle} \tilde{x}[n] e^{-jk(2\pi/N)n}. \quad (5.2)$$

Since $x[n] = \tilde{x}[n]$ over a period that includes the interval $-N_1 \leq n \leq N_2$, it is convenient to choose the interval of summation in eq. (5.2) to include this interval, so that $\tilde{x}[n]$ can be replaced by $x[n]$ in the summation. Therefore,

$$a_k = \frac{1}{N} \sum_{n=-N_1}^{N_2} x[n] e^{-jk(2\pi/N)n} = \frac{1}{N} \sum_{n=-\infty}^{+\infty} x[n] e^{-jk(2\pi/N)n}, \quad (5.3)$$

where in the second equality in eq. (5.3) we have used the fact that $x[n]$ is zero outside the interval $-N_1 \leq n \leq N_2$. Defining the function

$$X(e^{j\omega}) = \sum_{n=-\infty}^{+\infty} x[n] e^{-j\omega n}, \quad (5.4)$$

we see that the coefficients a_k are proportional to samples of $X(e^{j\omega})$, i.e.,

$$a_k = \frac{1}{N} X(e^{jk\omega_0}), \quad (5.5)$$

where $\omega_0 = 2\pi/N$ is the spacing of the samples in the frequency domain. Combining eqs. (5.1) and (5.5) yields

$$\tilde{x}[n] = \sum_{k=\langle N \rangle} \frac{1}{N} X(e^{jk\omega_0}) e^{jk\omega_0 n}. \quad (5.6)$$

Since $\omega_0 = 2\pi/N$, or equivalently, $1/N = \omega_0/2\pi$, eq. (5.6) can be rewritten as

$$\tilde{x}[n] = \frac{1}{2\pi} \sum_{k=\langle N \rangle} X(e^{jk\omega_0}) e^{jk\omega_0 n} \omega_0. \quad (5.7)$$

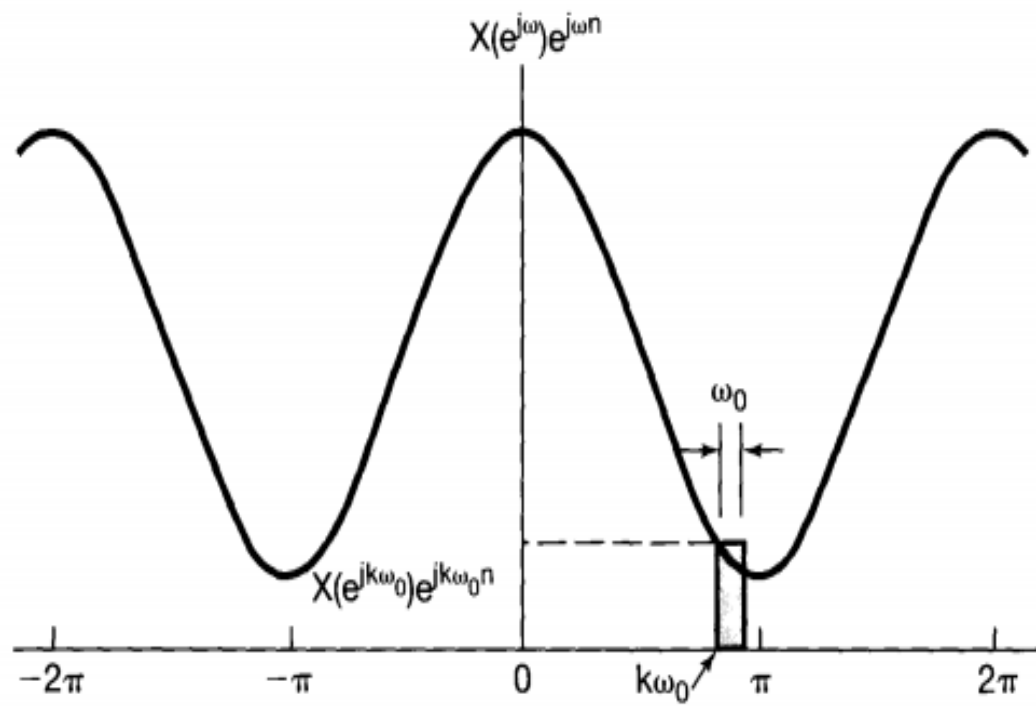


Figure 5.2 Graphical interpretation of eq. (5.7).

Synthesis and Analysis Equations

$$x[n] = \frac{1}{2\pi} \int_{2\pi} X(e^{j\omega}) e^{j\omega n} d\omega,$$

$$X(e^{j\omega}) = \sum_{n=-\infty}^{+\infty} x[n] e^{-j\omega n}.$$

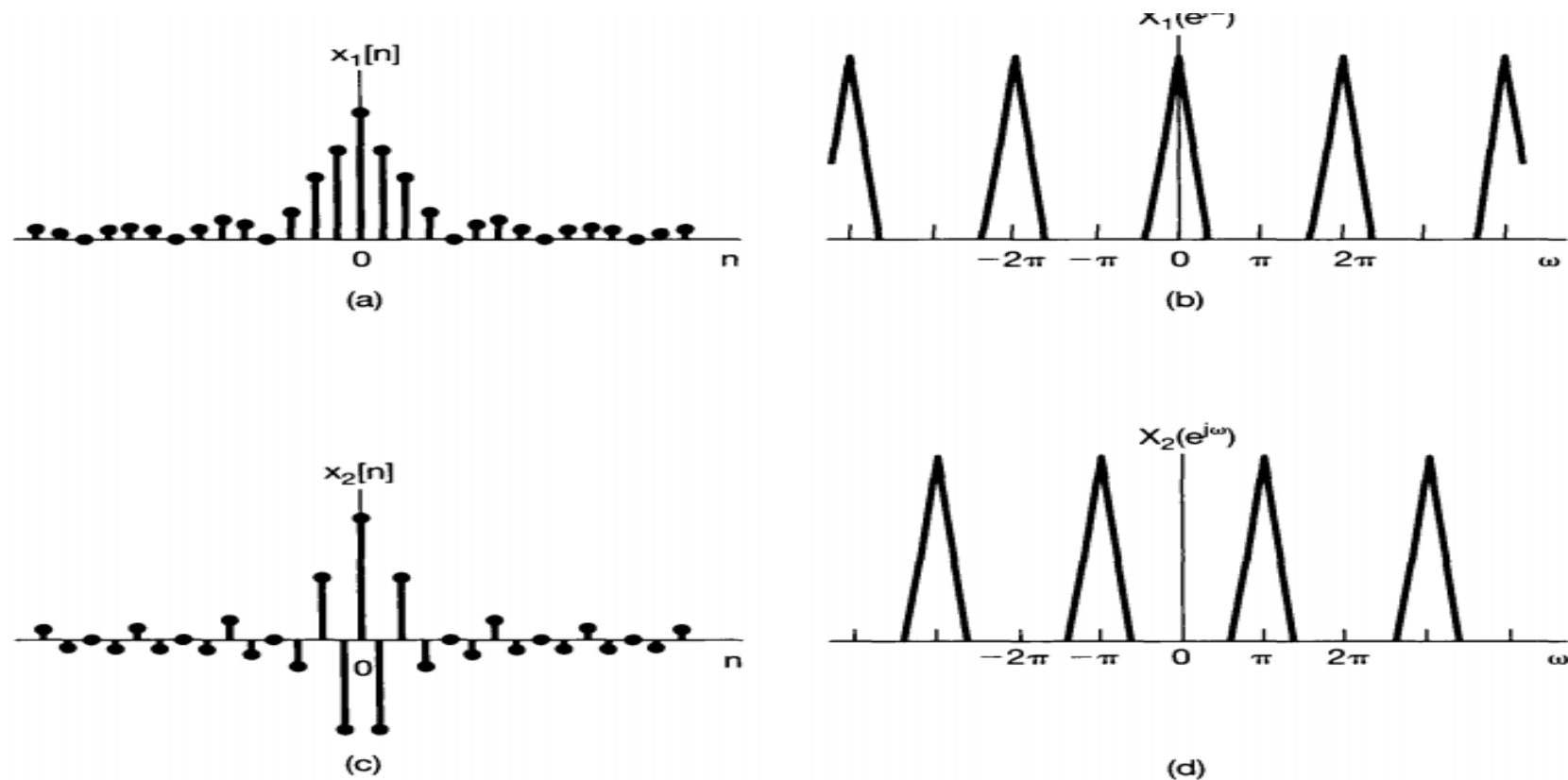


Figure 5.3 (a) Discrete-time signal $x_1[n]$. (b) Fourier transform of $x_1[n]$. Note that $X_1(e^{j\omega})$ is concentrated near $\omega = 0, \pm 2\pi, \pm 4\pi, \dots$. (c) Discrete-time signal $x_2[n]$. (d) Fourier transform of $x_2[n]$. Note that $X_2(e^{j\omega})$ is concentrated near $\omega = \pm\pi, \pm 3\pi, \dots$.

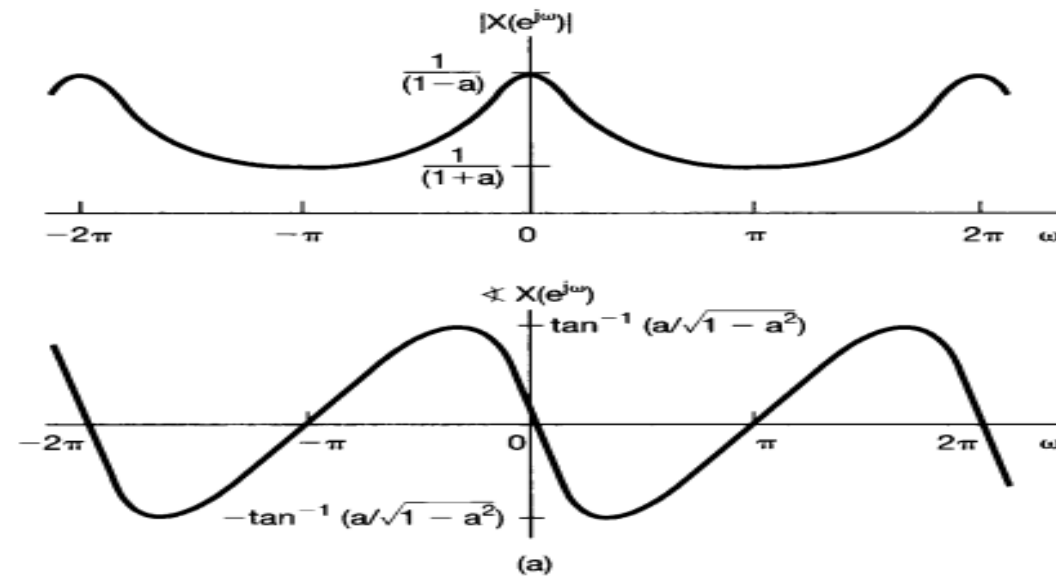
Consider the signal

$$x[n] = a^n u[n], \quad |a| < 1.$$

In this case,

$$\begin{aligned} X(e^{j\omega}) &= \sum_{n=-\infty}^{+\infty} a^n u[n] e^{-j\omega n} \\ &= \sum_{n=0}^{\infty} (ae^{-j\omega})^n = \frac{1}{1 - ae^{-j\omega}}. \end{aligned}$$

The magnitude and phase of $X(e^{j\omega})$ are shown in Figure 5.4(a) for $a > 0$ and in Figure 5.4(b) for $a < 0$. Note that all of these functions are periodic in ω with period 2π .



Fourier Transform of Periodic Signals

Now consider a periodic sequence $x[n]$ with period N and with the Fourier series representation

$$x[n] = \sum_{k=\langle N \rangle} a_k e^{jk(2\pi/N)n}. \quad (5.19)$$

In this case, the Fourier transform is

$$X(e^{j\omega}) = \sum_{k=-\infty}^{+\infty} 2\pi a_k \delta\left(\omega - \frac{2\pi k}{N}\right), \quad (5.20)$$

so that the Fourier transform of a periodic signal can be directly constructed from its Fourier coefficients.

TABLE 4.1 PROPERTIES OF THE FOURIER TRANSFORM

Section	Property	Aperiodic signal	Fourier transform
		$x(t)$ $y(t)$	$X(j\omega)$ $Y(j\omega)$
4.3.1	Linearity	$ax(t) + by(t)$	$aX(j\omega) + bY(j\omega)$
4.3.2	Time Shifting	$x(t - t_0)$	$e^{-j\omega t_0} X(j\omega)$
4.3.6	Frequency Shifting	$e^{j\omega_0 t} x(t)$	$X(j(\omega - \omega_0))$
4.3.3	Conjugation	$x^*(t)$	$X^*(-j\omega)$
4.3.5	Time Reversal	$x(-t)$	$X(-j\omega)$
4.3.5	Time and Frequency Scaling	$x(at)$	$\frac{1}{ a } X\left(\frac{j\omega}{a}\right)$
4.4	Convolution	$x(t) * y(t)$	$X(j\omega)Y(j\omega)$
4.5	Multiplication	$x(t)y(t)$	$\frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\theta)Y(j(\omega - \theta))d\theta$
4.3.4	Differentiation in Time	$\frac{d}{dt} x(t)$	$j\omega X(j\omega)$
4.3.4	Integration	$\int_{-\infty}^t x(t)dt$	$\frac{1}{j\omega} X(j\omega) + \pi X(0)\delta(\omega)$
4.3.6	Differentiation in Frequency	$tx(t)$	$j \frac{d}{d\omega} X(j\omega)$
4.3.3	Conjugate Symmetry for Real Signals	$x(t)$ real	$\begin{cases} X(j\omega) = X^*(-j\omega) \\ \Re\{X(j\omega)\} = \Re\{X(-j\omega)\} \\ \Im\{X(j\omega)\} = -\Im\{X(-j\omega)\} \\ X(j\omega) = X(-j\omega) \\ \angle X(j\omega) = -\angle X(-j\omega) \end{cases}$
4.3.3	Symmetry for Real and Even Signals	$x(t)$ real and even	$X(j\omega)$ real and even
4.3.3	Symmetry for Real and Odd Signals	$x(t)$ real and odd	$X(j\omega)$ purely imaginary and odd
4.3.3	Even-Odd Decomposition for Real Signals	$x_e(t) = \mathcal{E}v\{x(t)\}$ $[x(t) \text{ real}]$ $x_o(t) = \mathcal{O}d\{x(t)\}$ $[x(t) \text{ real}]$	$\Re\{X(j\omega)\}$ $j\Im\{X(j\omega)\}$
4.3.7	Parseval's Relation for Aperiodic Signals		
	$\int_{-\infty}^{+\infty} x(t) ^2 dt = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\omega) ^2 d\omega$		

SYSTEMS CHARACTERIZED BY LINEAR CONSTANT-COEFFICIENT DIFFERENTIAL EQUATIONS (Ex RLC)

$$\sum_{k=0}^N a_k \frac{d^k y(t)}{dt^k} = \sum_{k=0}^M b_k \frac{d^k x(t)}{dt^k}.$$

Consider an LTI system characterized by eq. (4.72). From the convolution property,

$$Y(j\omega) = H(j\omega)X(j\omega),$$

or equivalently,

$$H(j\omega) = \frac{Y(j\omega)}{X(j\omega)}, \quad (4.73)$$

where $X(j\omega)$, $Y(j\omega)$, and $H(j\omega)$ are the Fourier transforms of the input $x(t)$, output $y(t)$, and impulse response $h(t)$, respectively. Next, consider applying the Fourier transform to both sides of eq. (4.72) to obtain

$$\mathcal{F}\left\{\sum_{k=0}^N a_k \frac{d^k y(t)}{dt^k}\right\} = \mathcal{F}\left\{\sum_{k=0}^M b_k \frac{d^k x(t)}{dt^k}\right\}. \quad (4.74)$$

From the linearity property, eq. (4.26), this becomes

$$\sum_{k=0}^N a_k \mathcal{F}\left\{\frac{d^k y(t)}{dt^k}\right\} = \sum_{k=0}^M b_k \mathcal{F}\left\{\frac{d^k x(t)}{dt^k}\right\}, \quad (4.75)$$

and from the differentiation property, eq. (4.31),

$$\sum_{k=0}^N a_k(j\omega)^k Y(j\omega) = \sum_{k=0}^M b_k(j\omega)^k X(j\omega),$$

or equivalently,

$$Y(j\omega) \left[\sum_{k=0}^N a_k(j\omega)^k \right] = X(j\omega) \left[\sum_{k=0}^M b_k(j\omega)^k \right].$$

Thus, from eq. (4.73),

$$H(j\omega) = \frac{Y(j\omega)}{X(j\omega)} = \frac{\sum_{k=0}^M b_k(j\omega)^k}{\sum_{k=0}^N a_k(j\omega)^k}. \quad (4.76)$$

Example 4.24

Consider a stable LTI system characterized by the differential equation

$$\frac{dy(t)}{dt} + ay(t) = x(t), \quad (4.77)$$

with $a > 0$. From eq. (4.76), the frequency response is

$$H(j\omega) = \frac{1}{j\omega + a}. \quad (4.78)$$

Comparing this with the result of Example 4.1, we see that eq. (4.78) is the Fourier transform of $e^{-at}u(t)$. The impulse response of the system is then recognized as

$$h(t) = e^{-at}u(t).$$

Example 4.25

Consider a stable LTI system that is characterized by the differential equation

$$\frac{d^2 y(t)}{dt^2} + 4 \frac{dy(t)}{dt} + 3y(t) = \frac{dx(t)}{dt} + 2x(t).$$

From eq. (4.76), the frequency response is

$$H(j\omega) = \frac{(j\omega) + 2}{(j\omega)^2 + 4(j\omega) + 3}. \quad (4.79)$$

To determine the corresponding impulse response, we require the inverse Fourier transform of $H(j\omega)$. This can be found using the technique of partial-fraction expansion employed in Example 4.19 and discussed in detail in the appendix. (In particular, see Example A.1, in which the details of the calculations for the partial-fraction expansion of eq. (4.79) are worked out.) As a first step, we factor the denominator of the right-hand side of eq. (4.79) into a product of lower order terms:

$$H(j\omega) = \frac{j\omega + 2}{(j\omega + 1)(j\omega + 3)}. \quad (4.80)$$

Then, using the method of partial-fraction expansion, we find that

$$H(j\omega) = \frac{\frac{1}{2}}{j\omega + 1} + \frac{\frac{1}{2}}{j\omega + 3}.$$

The inverse transform of each term can be recognized from Example 4.24, with the result that

$$h(t) = \frac{1}{2}e^{-t}u(t) + \frac{1}{2}e^{-3t}u(t).$$

TABLE 5.1 PROPERTIES OF THE DISCRETE-TIME FOURIER TRANSFORM

Section	Property	Aperiodic Signal	Fourier Transform
		$x[n]$	$X(e^{j\omega})$ periodic with
		$y[n]$	$Y(e^{j\omega})$ period 2π
5.3.2	Linearity	$ax[n] + by[n]$	$aX(e^{j\omega}) + bY(e^{j\omega})$
5.3.3	Time Shifting	$x[n - n_0]$	$e^{-j\omega n_0} X(e^{j\omega})$
5.3.3	Frequency Shifting	$e^{j\omega_0 n} x[n]$	$X(e^{j(\omega - \omega_0)})$
5.3.4	Conjugation	$x^*[n]$	$X^*(e^{-j\omega})$
5.3.6	Time Reversal	$x[-n]$	$X(e^{-j\omega})$
5.3.7	Time Expansion	$x_{(k)}[n] = \begin{cases} x[n/k], & \text{if } n = \text{multiple of } k \\ 0, & \text{if } n \neq \text{multiple of } k \end{cases}$	$X(e^{jk\omega})$
5.4	Convolution	$x[n] * y[n]$	$X(e^{j\omega})Y(e^{j\omega})$
5.5	Multiplication	$x[n]y[n]$	$\frac{1}{2\pi} \int_{2\pi} X(e^{j\theta})Y(e^{j(\omega - \theta)})d\theta$
5.3.5	Differencing in Time	$x[n] - x[n - 1]$	$(1 - e^{-j\omega})X(e^{j\omega})$
5.3.5	Accumulation	$\sum_{k=-\infty}^n x[k]$	$\frac{1}{1 - e^{-j\omega}} X(e^{j\omega})$
			$+ \pi X(e^{j0}) \sum_{k=-\infty}^{+\infty} \delta(\omega - 2\pi k)$
5.3.8	Differentiation in Frequency	$nx[n]$	$j \frac{dX(e^{j\omega})}{d\omega}$
5.3.4	Conjugate Symmetry for Real Signals	$x[n]$ real	$\begin{cases} X(e^{j\omega}) = X^*(e^{-j\omega}) \\ \Re\{X(e^{j\omega})\} = \Re\{X(e^{-j\omega})\} \\ \Im\{X(e^{j\omega})\} = -\Im\{X(e^{-j\omega})\} \\ X(e^{j\omega}) = X(e^{-j\omega}) \\ \angle X(e^{j\omega}) = -\angle X(e^{-j\omega}) \end{cases}$
5.3.4	Symmetry for Real, Even Signals	$x[n]$ real and even	$X(e^{j\omega})$ real and even
5.3.4	Symmetry for Real, Odd	$x[n]$ real and odd	$X(e^{j\omega})$ purely imaginary and

SYSTEMS CHARACTERIZED BY LINEAR CONSTANT-COEFFICIENT DIFFERENCE EQUATIONS

A general linear constant-coefficient difference equation for an LTI system with input $x[n]$ and output $y[n]$ is of the form

$$\sum_{k=0}^N a_k y[n-k] = \sum_{k=0}^M b_k x[n-k]. \quad (5.78)$$

$$H(e^{j\omega}) = \frac{Y(e^{j\omega})}{X(e^{j\omega})}.$$

Applying the Fourier transform to both sides of eq. (5.78) and using the linearity and time-shifting properties, we obtain the expression

$$\sum_{k=0}^N a_k e^{-jk\omega} Y(e^{j\omega}) = \sum_{k=0}^M b_k e^{-jk\omega} X(e^{j\omega}),$$

or equivalently,

$$H(e^{j\omega}) = \frac{Y(e^{j\omega})}{X(e^{j\omega})} = \frac{\sum_{k=0}^M b_k e^{-jk\omega}}{\sum_{k=0}^N a_k e^{-jk\omega}}. \quad (5.80)$$

Example 5.18

Consider the causal LTI system that is characterized by the difference equation

$$y[n] - ay[n - 1] = x[n], \quad (5.81)$$

with $|a| < 1$. From eq. (5.80), the frequency response of this system is

$$H(e^{j\omega}) = \frac{1}{1 - ae^{-j\omega}}. \quad (5.82)$$

Comparing this with Example 5.1, we recognize it as the Fourier transform of the sequence $a^n u[n]$. Thus, the impulse response of the system is

$$h[n] = a^n u[n]. \quad (5.83)$$

Example 5.19

Consider a causal LTI system that is characterized by the difference equation

$$y[n] - \frac{3}{4}y[n-1] + \frac{1}{8}y[n-2] = 2x[n]. \quad (5.84)$$

From eq. (5.80), the frequency response is

$$H(e^{j\omega}) = \frac{2}{1 - \frac{3}{4}e^{-j\omega} + \frac{1}{8}e^{-j2\omega}}. \quad (5.85)$$

As a first step in obtaining the impulse response, we factor the denominator of eq. (5.85):

$$H(e^{j\omega}) = \frac{2}{(1 - \frac{1}{2}e^{-j\omega})(1 - \frac{1}{4}e^{-j\omega})}. \quad (5.86)$$

$H(e^{j\omega})$ can be expanded by the method of partial fractions, as in Example A.3 in the appendix. The result of this expansion is

$$H(e^{j\omega}) = \frac{4}{1 - \frac{1}{2}e^{-j\omega}} - \frac{2}{1 - \frac{1}{4}e^{-j\omega}}. \quad (5.87)$$

The inverse transform of each term can be recognized by inspection, with the result that

$$h[n] = 4\left(\frac{1}{2}\right)^n u[n] - 2\left(\frac{1}{4}\right)^n u[n]. \quad (5.88)$$

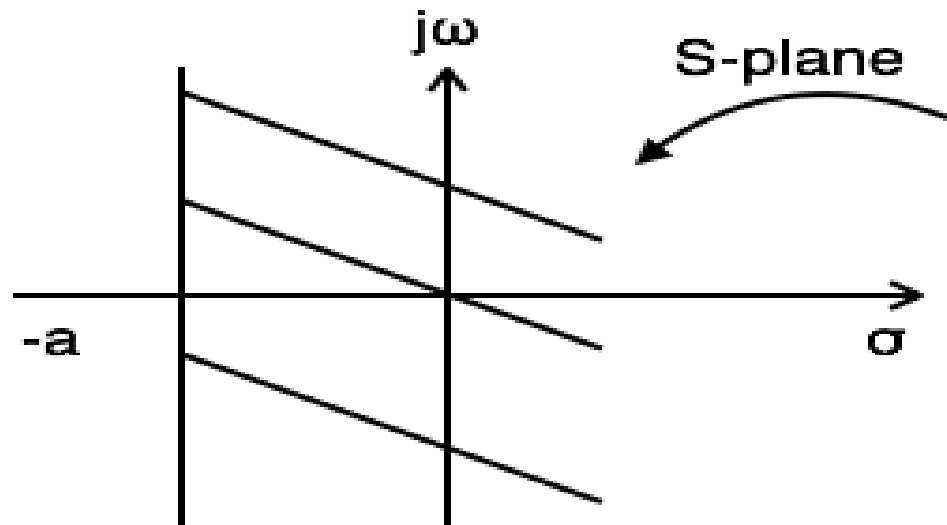
LT

Example 1: Find the Laplace transform and ROC $x(t)=e(-at)u(t)$

$$\text{L.T}[x(t)]=\text{L.T}[e(-at)u(t)]=1/(s+a)$$

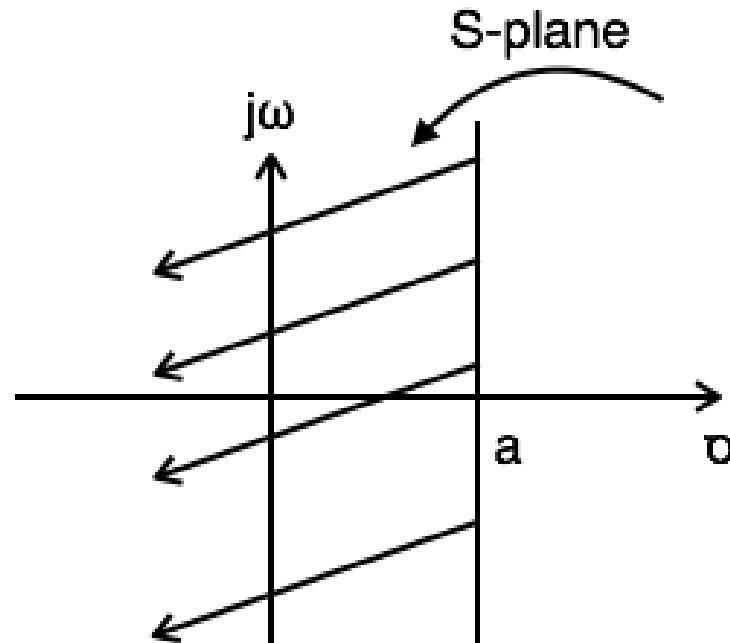
$$\text{Re}(s)>-a$$

$$\text{ROC: Re}(s)>>-a$$



Example 2: Find the Laplace transform and ROC of $x(t)=e(at)u(-t)$

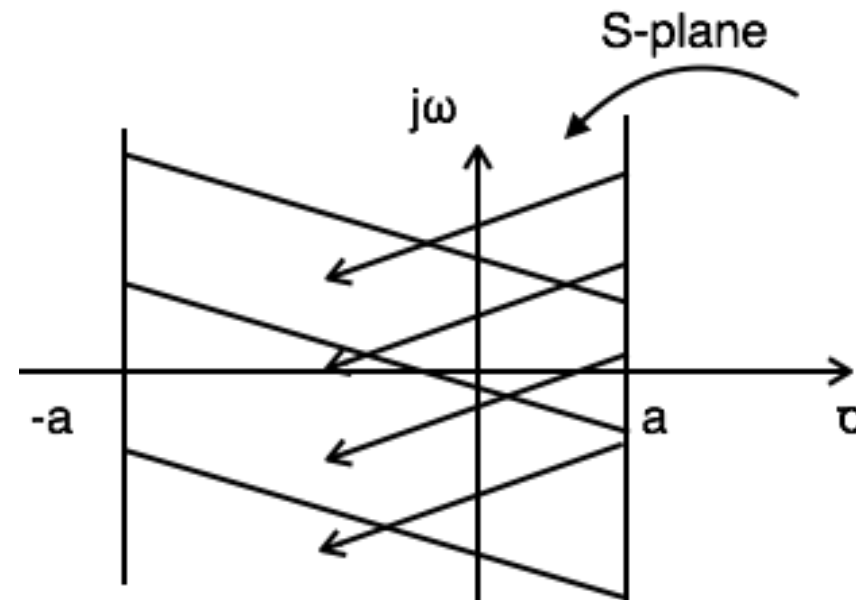
- $L.T[x(t)]=L.T[e(at)u(-t)]=1/(s-a)$
- $\text{Re}(s)<a$
- ROC: $\text{Re}(s)<a$



Find the Laplace transform and ROC of $x(t)=e(-at)u(t)+e(at)u(-t)$

- $L.T[x(t)]=L.T[e(-at)u(t)+e(at)u(-t)]=1/(s+a)+1/(s-a)$
- For $1/(s+a)$ $\text{Re}\{s\}>-a$
- For $1/(s-a)$ $\text{Re}\{s\}<a$

ROC: $-a < \text{Re}(s) < a$



■ CAUSALITY AND STABILITY

The impulse response is the inverse Laplace transform of the transfer function. In order to obtain a unique inverse transform, we must know the ROC or have other knowledge of the impulse response. The differential-equation description for a system does not contain this information. Hence to obtain the impulse response we must have additional knowledge of the system characteristics. The relationships between the poles, zeros, and system characteristics can provide this additional knowledge.

The impulse response of a causal system is zero for $t < 0$. Therefore if we know that a system is causal, the impulse response is determined from the transfer function by using

right-sided inverse Laplace transforms. A pole at $s = d_k$ in the left half of the s -plane [$\text{Re}(d_k) < 0$] contributes an exponentially decaying term to the impulse response, while a pole in the right half of the s -plane [$\text{Re}(d_k) > 0$] contributes an increasing exponential term to the impulse response. These relationships are illustrated in Fig. 6.17.

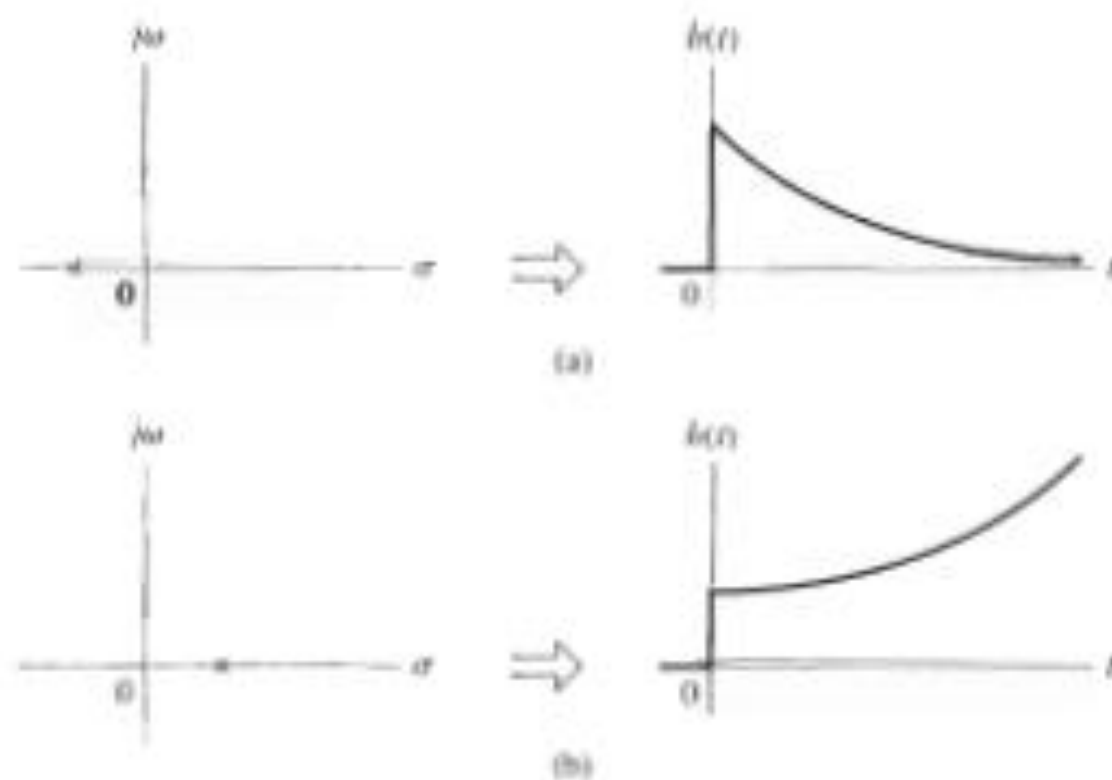


FIGURE 6.17 Relationship between the pole location and impulse response for a causal system. (a) A pole in the left half of the s -plane corresponds to an exponentially decaying impulse response. (b) A pole in the right half of the s -plane corresponds to an exponentially increasing impulse response.

Alternatively, if we know a system is stable, then the impulse response is absolutely integrable. This implies that the Fourier transform exists and thus the ROC includes the $j\omega$ -axis in the s -plane. This knowledge is sufficient to uniquely determine the inverse La-

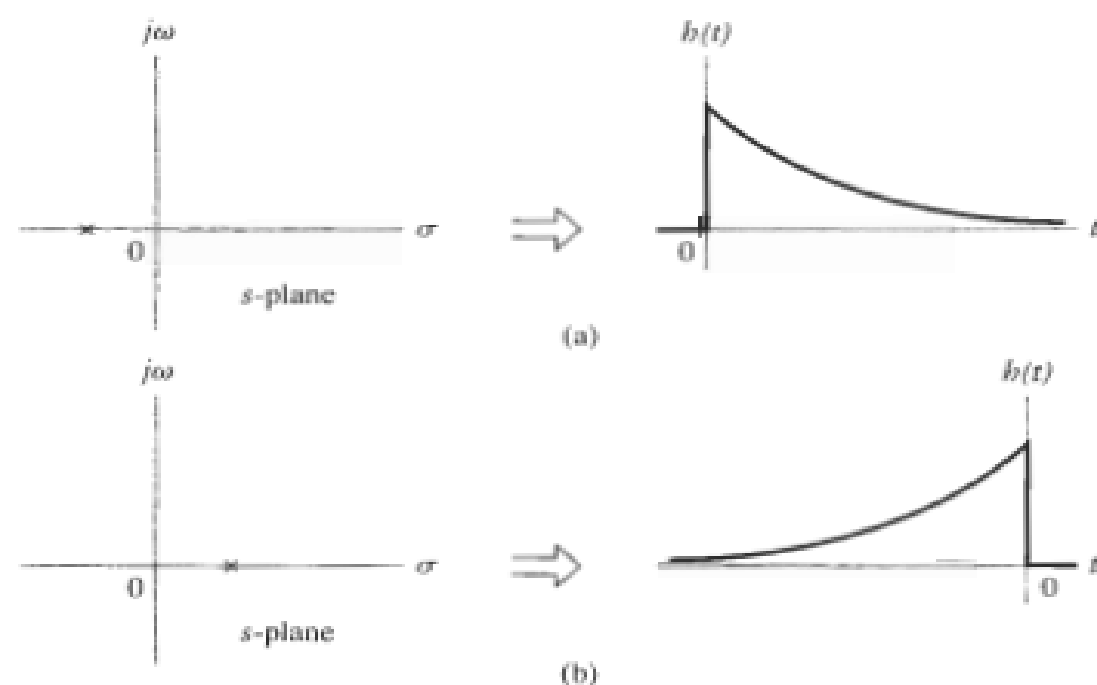


FIGURE 6.18 Relationship between the pole location and impulse response for a stable system. (a) A pole in the left half of the s -plane corresponds to a right-sided impulse response. (b) A pole in the right half of the s -plane corresponds to a left-sided impulse response.

place transform of the transfer function. A pole that is in the right half of the s -plane contributes a left-sided decaying exponential term to the impulse response, while a pole in the left half of the s -plane contributes a right-sided decaying exponential term to the impulse response as depicted in Fig. 6.18. Note that a stable impulse response cannot contain any increasing exponential terms, since an increasing exponential is not absolutely integrable.

Suppose a system is known to be both causal and stable. A pole that is in the left half of the s -plane contributes a right-sided decaying exponential term to the impulse response. We cannot have a pole in the right half of the s -plane, however. A pole in the right half will contribute either a left-sided decaying exponential that is not causal, or a right-sided increasing exponential that results in an unstable impulse response. That is, the inverse Laplace transform of a pole in the right half of the s -plane is either stable or causal, but cannot be both stable and causal. Systems that are stable and causal must have all their poles in the left half of the s -plane. Such a system is illustrated in Fig. 6.19.

Stable and causal System

- All poles must be in left half of the s-plane.
- If a system is only causal then $h(t) = 0$ for $t < 0$, i.e. $h(t)$ must be a right sided function or sequence as shown in Fig. 6.17.
- If a system is only stable its impulse response must be absolutely integrable, so poles must be in left half of the s-plane.

EXAMPLE 6.19 A system has the transfer function

$$H(s) = \frac{2}{s+3} + \frac{1}{s-2}$$

Find the impulse response assuming (a) the system is stable and (b) the system is causal. Can this system be both stable and causal?

Solution: This system has a pole at $s = -3$ and at $s = 2$. If the system is stable, then the pole at $s = -3$ contributes a right-sided term to the impulse response, while the pole at $s = 2$ contributes a left-sided term. We have

$$h(t) = 2e^{-3t}u(t) - e^{2t}u(-t)$$

If the system is causal, then both poles must contribute right-sided terms to the impulse response and we have

$$h(t) = 2e^{-3t}u(t) + e^{2t}u(t)$$

Note that this system is not stable since the term $e^{2t}u(t)$ is not absolutely integrable. In fact, this system cannot be both stable and causal since the pole at $s = 2$ is in the right half of the s -plane.

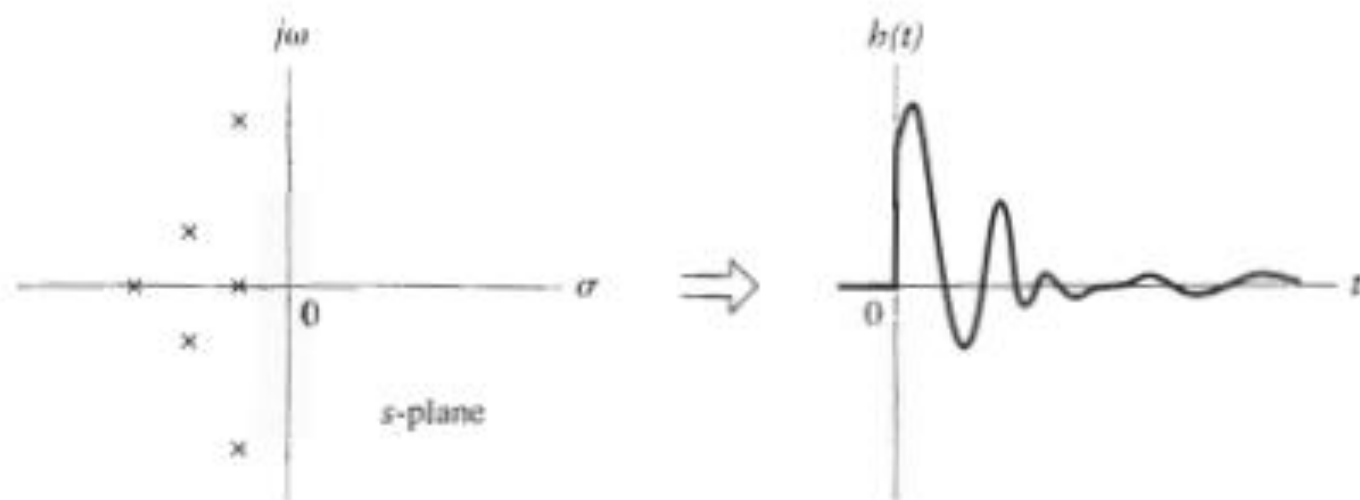


FIGURE 6.19 A system that is both stable and causal must have all of its poles in the left half of the s -plane, as shown here.

■ INVERSE SYSTEMS

Given a LTI system with impulse response $h(t)$, the impulse response of the inverse system, $h^{-1}(t)$, satisfies the condition

$$h^{-1}(t) * h(t) = \delta(t)$$

If we take the Laplace transform of both sides of this equation, we find that the inverse system transfer function $H^{-1}(s)$ satisfies

$$H^{-1}(s)H(s) = 1$$

or

$$H^{-1}(s) = \frac{1}{H(s)}$$

The inverse system's transfer function is therefore the inverse of the transfer function of the original system. If $H(s)$ is written in pole-zero form, as in Eq. (6.31), then we have

$$H^{-1}(s) = \frac{\prod_{k=1}^N (s - d_k)}{(b_M/a_N) \prod_{k=1}^M (s - c_k)} \quad (6.36)$$

The zeros of the inverse system are the poles of $H(s)$, and the poles of the inverse system are the zeros of $H(s)$. We conclude that any system with a rational transfer function has an inverse system.

EXAMPLE 6.20 Consider a system H described by the differential equation

$$\frac{d}{dt} y(t) + 2y(t) = \frac{d^2}{dt^2} x(t) + 2 \frac{d}{dt} x(t) - 3x(t)$$

Find the transfer function of the inverse system for H . Does a stable and causal inverse system exist?

Solution: First find the system transfer function $H(s)$ by taking the Laplace transform of both sides of the given differential equation, obtaining

$$Y(s)(s + 2) = X(s)(s^2 + 2s - 3)$$

Hence the transfer function of the system is

$$\begin{aligned} H(s) &= \frac{Y(s)}{X(s)} \\ &= \frac{s^2 + 2s - 3}{s + 2} \end{aligned}$$

and the inverse system has the transfer function

$$\begin{aligned} H^{-1}(s) &= \frac{1}{H(s)} \\ &= \frac{s + 2}{s^2 + 2s - 3} \\ &= \frac{s + 2}{(s - 1)(s + 3)} \end{aligned}$$

The inverse system has poles at $s = 1$ and $s = -3$. The pole at $s = 1$ is in the right half of the s -plane. Therefore the inverse system cannot be both stable and causal.

Solving differential equations

The general form for the differentiation property is

$$\begin{aligned} \frac{d^n}{dt^n} x(t) \xleftrightarrow{\mathcal{L}_u} s^n X(s) - \frac{d^{n-1}}{dt^{n-1}} x(t) \Big|_{t=0^+} - s \frac{d^{n-2}}{dt^{n-2}} x(t) \Big|_{t=0^+} \\ - \dots - s^{n-2} \frac{d}{dt} x(t) \Big|_{t=0^+} - s^{n-1} x(0^+) \end{aligned} \quad (6.17)$$

Integration Property

$$\int_{-\infty}^t x(\tau) d\tau \xleftrightarrow{\mathcal{L}_u} \frac{x^{(-1)}(0^+)}{s} + \frac{X(s)}{s} \quad (6.18)$$

where

$$x^{(-1)}(0^+) = \int_{-\infty}^{0^+} x(\tau) d\tau$$

is the area under $x(t)$ from $t = -\infty$ to $t = 0^+$,

Initial and Final Value Theorems

The initial and final value theorems allow us to determine the initial value of $x(t)$, $x(0^+)$, and the final value, $x(\infty)$, directly from $X(s)$. These theorems are most often used to evaluate either the initial or final values of a system output without explicitly determining the entire time response of the system. The initial value theorem states that

$$\lim_{s \rightarrow \infty} sX(s) = x(0^+) \quad (6.19)$$

The initial value theorem does not apply to rational functions $X(s)$ whose numerator polynomial order is greater than the denominator polynomial order. The final value theorem states that

$$\lim_{s \rightarrow 0} sX(s) = x(\infty) \quad (6.20)$$

The final value theorem applies only if all the poles of $X(s)$ are in the left half of the s -plane, with at most a single pole at $s = 0$.

EXAMPLE 6.5 Determine the initial and final values of a signal $x(t)$ whose unilateral Laplace transform is

$$X(s) = \frac{7s + 10}{s(s + 2)}$$

Solution: We may apply the initial value theorem to obtain

$$\begin{aligned}x(0^+) &= \lim_{s \rightarrow \infty} s \frac{7s + 10}{s(s + 2)} \\&= \lim_{s \rightarrow \infty} \frac{7s + 10}{s + 2} \\&= 7\end{aligned}$$

The final value theorem is applicable since $X(s)$ has only a single pole at $s = 0$ and the remaining poles are in the left half of the s -plane. We have

$$\begin{aligned}x(\infty) &= \lim_{s \rightarrow 0} s \frac{7s + 10}{s(s + 2)} \\&= \lim_{s \rightarrow 0} \frac{7s + 10}{s + 2} \\&= 5\end{aligned}$$

The reader may verify these results by showing the inverse Laplace transform of $X(s)$ is $x(t) = 5u(t) + 2e^{-2t}u(t)$.

EXAMPLE 6.6 Find the inverse Laplace transform of

$$X(s) = \frac{3s + 4}{(s + 1)(s + 2)^2}$$

Solution: Use a partial fraction expansion of $X(s)$ to write

$$X(s) = \frac{A_1}{s + 1} + \frac{A_2}{s + 2} + \frac{A_3}{(s + 2)^2}$$

Solving for A_1 , A_2 , and A_3 using the method of residues, we obtain

$$X(s) = \frac{1}{s + 1} - \frac{1}{s + 2} + \frac{2}{(s + 2)^2}$$

We may construct $x(t)$ from the inverse Laplace transform of each term in the partial fraction expansion:

- The pole of the first term is at $s = -1$:

$$e^{-t}u(t) \xleftrightarrow{\mathcal{L}_u} \frac{1}{s+1}$$

- The second term has a pole at $s = -2$:

$$-e^{-2t}u(t) \xleftrightarrow{\mathcal{L}_u} -\frac{1}{s+2}$$

- The double pole in the last term is at $s = -2$:

$$2te^{-2t}u(t) \xleftrightarrow{\mathcal{L}_u} \frac{2}{(s+2)^2}$$

Combining these three terms, we obtain

$$x(t) = e^{-t}u(t) - e^{-2t}u(t) + 2te^{-2t}u(t)$$

EXAMPLE 6.8 Use the Laplace transform to find the output of the system described by the differential equation

$$\frac{d}{dt}y(t) + 5y(t) = x(t)$$

in response to the input $x(t) = 3e^{-2t}u(t)$ and initial condition $y(0^+) = -2$.

Solution: Take the unilateral Laplace transform of each side of the differential equation and apply the differentiation property of Eq. (6.16) to obtain

$$sY(s) - y(0^+) + 5Y(s) = X(s)$$

Solving for $Y(s)$, we get

$$Y(s) = \frac{1}{s+5} [X(s) + y(0^+)]$$

Now substitute $X(s) = 3/(s+2)$ and the initial condition $y(0^+) = -2$, obtaining

$$Y(s) = \frac{3}{(s+2)(s+5)} + \frac{-2}{s+5}$$

Expanding $Y(s)$ in partial fractions,

$$Y(s) = \frac{1}{s+2} + \frac{-1}{s+5} + \frac{-2}{s+5}$$

and taking the inverse unilateral Laplace transform, we get

$$y(t) = e^{-2t}u(t) - 3e^{-5t}u(t)$$

EXAMPLE 6.9 Use the Laplace transform to determine the output of a system represented by the differential equation

$$\frac{d^2}{dt^2} y(t) + 5 \frac{d}{dt} y(t) + 6y(t) = 2 \frac{d}{dt} x(t) + x(t)$$

in response to the input $x(t) = u(t)$. Assume that the initial conditions on the system are

$$y(0^+) = 1 \quad \text{and} \quad \frac{d}{dt} y(t)|_{t=0^+} = 2$$

Identify the forced response of the system, $y^{(f)}(t)$, and the natural response, $y^{(n)}(t)$.

Solution: Taking the unilateral Laplace transform of both sides of the differential equation, we obtain

$$(s^2 + 5s + 6)Y(s) - sy(0^+) - \frac{d}{dt} y(t)|_{t=0^+} - 5y(0^+) = (2s + 1)X(s) - 2x(0^+)$$

Solving for $Y(s)$, we get

$$Y(s) = \frac{(2s + 1)X(s) - 2x(0^+)}{s^2 + 5s + 6} + \frac{sy(0^+) + \frac{d}{dt}y(t)|_{t=0^+} + 5y(0^+)}{s^2 + 5s + 6}$$

The first term is associated with the forced response of the system, $Y^{(f)}(s)$. The second term corresponds to the natural response, $Y^{(n)}(s)$. Substituting for $X(s)$, $x(0^+)$, and the initial conditions, we obtain

$$Y^{(f)}(s) = \frac{1}{s(s + 2)(s + 3)}$$

and

$$Y^{(n)}(s) = \frac{s + 7}{(s + 2)(s + 3)}$$

Expanding both of these in a partial fraction expansion yields

$$Y^{(f)}(s) = \frac{\frac{1}{6}}{s} + \frac{-\frac{1}{2}}{s + 2} + \frac{\frac{1}{3}}{s + 3}$$

$$Y^{(n)}(s) = \frac{5}{s + 2} + \frac{-4}{s + 3}$$

Next, taking the inverse unilateral Laplace transforms of $Y^{(f)}(s)$ and $Y^{(n)}(s)$, we obtain

$$y^{(f)}(t) = \frac{1}{6}u(t) - \frac{1}{2}e^{-2t}u(t) + \frac{1}{3}e^{-3t}u(t)$$

and

$$y^{(n)}(t) = 5e^{-2t}u(t) - 4e^{-3t}u(t)$$

The output of the system is $y(t) = y^{(f)}(t) + y^{(n)}(t)$. Figures 6.7(a), (b), and (c) depict the forced response, natural response, and system output, respectively.

The first term, $Y^{(f)}(s)$, represents the component of the response associated entirely with the input, or the *forced response* of the system. It represents the output when the initial conditions are zero. The second term, $Y^{(n)}(s)$, represents the component of the output due to the initial conditions, or the *natural response* of the system. It represents the system output when the input is zero.

TABLE 9.3 PROPERTIES OF THE UNILATERAL LAPLACE TRANSFORM

Property	Signal	Unilateral Laplace Transform
	$x(t)$ $x_1(t)$ $x_2(t)$	$\mathfrak{X}(s)$ $\mathfrak{X}_1(s)$ $\mathfrak{X}_2(s)$
Linearity	$ax_1(t) + bx_2(t)$	$a\mathfrak{X}_1(s) + b\mathfrak{X}_2(s)$
Shifting in the s -domain	$e^{s_0 t} x(t)$	$\mathfrak{X}(s - s_0)$
Time scaling	$x(at), \quad a > 0$	$\frac{1}{a} \mathfrak{X}\left(\frac{s}{a}\right)$
Conjugation	$x^*(t)$	$x^*(s)$
Convolution (assuming that $x_1(t)$ and $x_2(t)$ are identically zero for $t < 0$)	$x_1(t) * x_2(t)$	$\mathfrak{X}_1(s) \mathfrak{X}_2(s)$
Differentiation in the time domain	$\frac{d}{dt} x(t)$	$s \mathfrak{X}(s) - x(0^-)$
Differentiation in the s -domain	$-tx(t)$	$\frac{d}{ds} \mathfrak{X}(s)$
Integration in the time domain	$\int_{0^-}^t x(\tau) d\tau$	$\frac{1}{s} \mathfrak{X}(s)$

Initial- and Final-Value Theorems

If $x(t)$ contains no impulses or higher-order singularities at $t = 0$, then

$$x(0^+) = \lim_{s \rightarrow \infty} s \mathfrak{X}(s)$$

$$\lim_{t \rightarrow \infty} x(t) = \lim_{s \rightarrow 0} s \mathfrak{X}(s)$$

Z Transform

$$X(s) = \int_{-\infty}^{\infty} x(t) e^{-st} dt \stackrel{t=n\Delta t}{=} \lim_{\Delta t \rightarrow 0} \sum_{n=-\infty}^{\infty} \underbrace{x(n\Delta t) \Delta t}_{x[n]} \underbrace{\left(e^{s\Delta t}\right)^{-n}}_z \Rightarrow X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n}$$

$$z = e^{s\Delta t} = e^{j\omega\Delta t} \Rightarrow |z| = |e^{j\omega\Delta t}| = 1 \Rightarrow j\omega \text{ - axis maps to the unit circle}$$

Hence, we have shown that complex exponentials are eigenfunctions of LTI systems. The constant $H(s)$ for a specific value of s is then the eigenvalue associated with the eigenfunction e^{st} .

In an exactly parallel manner, we can show that complex exponential sequences are eigenfunctions of discrete-time LTI systems. That is, suppose that an LTI system with impulse response $h[n]$ has as its input the sequence

$$x[n] = z^n, \quad (3.7)$$

where z is a complex number. Then the output of the system can be determined from the convolution sum as

$$\begin{aligned} y[n] &= \sum_{k=-\infty}^{+\infty} h[k]x[n-k] \\ &= \sum_{k=-\infty}^{+\infty} h[k]z^{n-k} = z^n \sum_{k=-\infty}^{+\infty} h[k]z^{-k}. \end{aligned} \quad (3.8)$$

From this expression, we see that if the input $x[n]$ is the complex exponential given by eq. (3.7), then, assuming that the summation on the right-hand side of eq. (3.8) converges, the output is the same complex exponential multiplied by a constant that depends on the

value of z . That is,

$$y[n] = H(z)z^n, \quad (3.9)$$

where

$$H(z) = \sum_{k=-\infty}^{+\infty} h[k]z^{-k}. \quad (3.10)$$

Consequently, as in the continuous-time case, complex exponentials are eigenfunctions of discrete-time LTI systems. The constant $H(z)$ for a specified value of z is the eigenvalue associated with the eigenfunction z^n .

Z-Transform—Discrete time counterpart of Laplace Transform

- Motivation and properties are parallel to the Laplace transform

As we saw in Section 3.2, for a discrete-time linear time-invariant system with impulse response $h[n]$, the response $y[n]$ of the system to a complex exponential input of the form z^n is

$$y[n] = H(z)z^n, \quad (10.1)$$

$$H(z) = \sum_{n=-\infty}^{+\infty} h[n]z^{-n}.$$

For $z = e^{j\omega}$ with ω real (i.e., with $|z| = 1$),

The z -transform of a general discrete-time signal $x[n]$ is defined as¹

$$X(z) \triangleq \sum_{n=-\infty}^{+\infty} x[n]z^{-n}, \quad (10.3)$$

where z is a complex variable. For convenience, the z -transform of $x[n]$ will sometimes be denoted as $\mathcal{Z}\{x[n]\}$ and the relationship between $x[n]$ and its z -transform indicated as

$$x[n] \xleftrightarrow{\mathcal{Z}} X(z). \quad (10.4)$$

Relationship between Z Transform and DTFT

$$z = re^{j\omega}, \quad (10.5)$$

with r as the magnitude of z and ω as the angle of z . In terms of r and ω , eq. (10.3) becomes

$$X(re^{j\omega}) = \sum_{n=-\infty}^{+\infty} x[n](re^{j\omega})^{-n},$$

or equivalently,

$$X(re^{j\omega}) = \sum_{n=-\infty}^{+\infty} \{x[n]r^{-n}\}e^{-j\omega n}. \quad (10.6)$$

From eq. (10.6), we see that $X(re^{j\omega})$ is the Fourier transform of the sequence $x[n]$ multiplied by a real exponential r^{-n} ; that is,

$$X(re^{j\omega}) = \mathcal{F}\{x[n]r^{-n}\}. \quad (10.7)$$

The exponential weighting r^{-n} may be decaying or growing with increasing n , depending on whether r is greater than or less than unity. We note in particular that, for $r = 1$, or equivalently, $|z| = 1$, eq. (10.3) reduces to the Fourier transform; that is,

$$X(z) \Big|_{z=e^{j\omega}} = X(e^{j\omega}) = \mathcal{F}\{x[n]\}. \quad (10.8)$$

The relationship between the z -transform and Fourier transform for discrete-time signals parallels closely the corresponding discussion in Section 9.1 for continuous-time signals, but with some important differences. In the continuous-time case, the Laplace transform reduces to the Fourier transform when the real part of the transform variable is zero. Interpreted in terms of the s -plane, this means that the Laplace transform reduces to the Fourier transform on the imaginary axis (i.e., for $s = j\omega$). In contrast, the z -transform reduces to the Fourier transform when the magnitude of the transform variable z is *unity* (i.e., for $z = e^{j\omega}$). Thus, the z -transform reduces to the Fourier transform on the contour in the complex z -plane corresponding to a circle with a radius of unity, as indicated in Figure 10.1. This circle in the z -plane is referred to as the *unit circle* and plays a role in the discussion of the z -transform similar to the role of the imaginary axis in the s -plane for the Laplace transform.

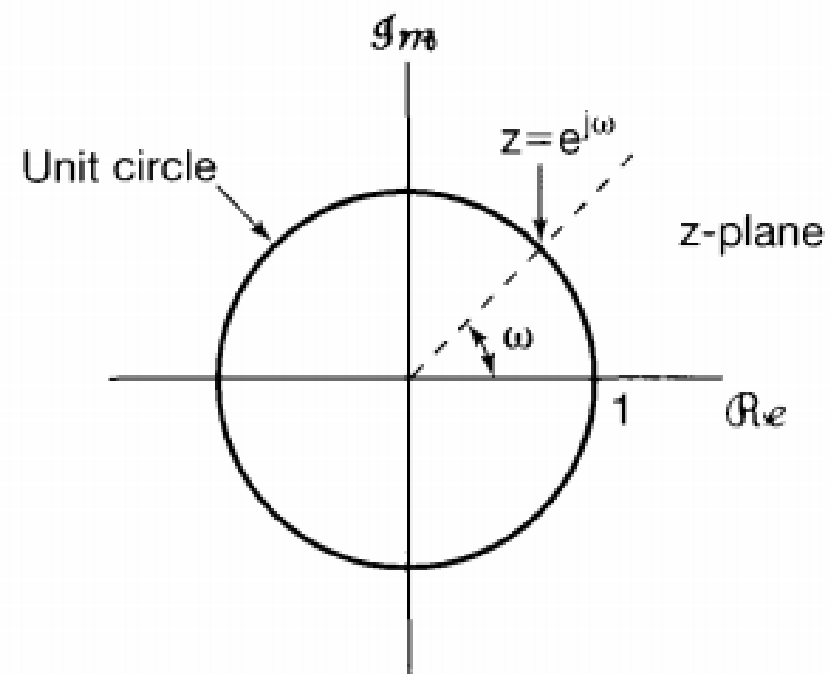


Figure 10.1 Complex z -plane. The z -transform reduces to the Fourier transform for values of z on the unit circle.

From eq. (10.7), we see that, for convergence of the z -transform, we require that the Fourier transform of $x[n]r^{-n}$ converge. For any specific sequence $x[n]$, we would expect this convergence for some values of r and not for others. In general, the z -transform of a sequence has associated with it a range of values of z for which $X(z)$ converges. As with the Laplace transform, this range of values is referred to as the *region of convergence* (ROC). If the ROC includes the unit circle, then the Fourier transform also converges. To illustrate the z -transform and the associated region of convergence, let us consider several examples.

Consider the signal $x[n] = a^n u[n]$. Then, from eq. (10.3),

$$X(z) = \sum_{n=-\infty}^{+\infty} a^n u[n] z^{-n} = \sum_{n=0}^{\infty} (az^{-1})^n.$$

For convergence of $X(z)$, we require that $\sum_{n=0}^{\infty} |az^{-1}|^n < \infty$. Consequently, the region of convergence is the range of values of z for which $|az^{-1}| < 1$, or equivalently, $|z| > |a|$.

Then

$$X(z) = \sum_{n=0}^{\infty} (az^{-1})^n = \frac{1}{1 - az^{-1}} = \frac{z}{z - a}, \quad |z| > |a|. \quad (10.9)$$

Thus, the z -transform for this signal is well-defined for any value of a , with an ROC determined by the magnitude of a according to eq. (10.9). For example, for $a = 1$, $x[n]$ is the unit step sequence with z -transform

$$X(z) = \frac{1}{1 - z^{-1}}, \quad |z| > 1.$$

We see that the z -transform in eq. (10.9) is a rational function. Consequently, just as with rational Laplace transforms, the z -transform can be characterized by its zeros (the roots of the numerator polynomial) and its poles (the roots of the denominator polynomial). For this example, there is one zero, at $z = 0$, and one pole, at $z = a$. The pole-zero plot and the region of convergence for Example 10.1 are shown in Figure 10.2 for a value of a between 0 and 1. For $|a| > 1$, the ROC does not include the unit circle, consistent with the fact that, for these values of a , the Fourier transform of $a^n u[n]$ does not converge.

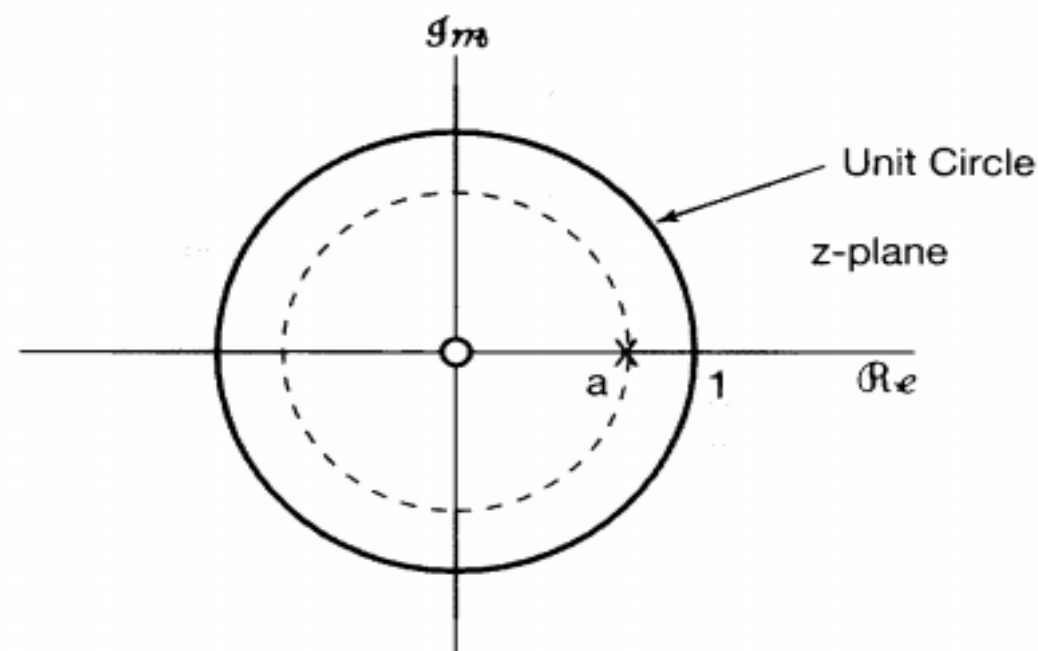


Figure 10.2 Pole-zero plot and region of convergence for Example 10.1 for $0 < a < 1$.

Example 10.2

Now let $x[n] = -a^n u[-n - 1]$. Then

$$\begin{aligned} X(z) &= - \sum_{n=-\infty}^{+\infty} a^n u[-n - 1] z^{-n} = - \sum_{n=-\infty}^{-1} a^n z^{-n} \\ &= - \sum_{n=1}^{\infty} a^{-n} z^n = 1 - \sum_{n=0}^{\infty} (a^{-1} z)^n. \end{aligned} \quad (10.10)$$

If $|a^{-1} z| < 1$, or equivalently, $|z| < |a|$, the sum in eq. (10.10) converges and

$$X(z) = 1 - \frac{1}{1 - a^{-1} z} = \frac{1}{1 - a z^{-1}} = \frac{z}{z - a}, \quad |z| < |a|. \quad (10.11)$$

The pole-zero plot and region of convergence for this example are shown in Figure 10.3 for a value of a between 0 and 1.

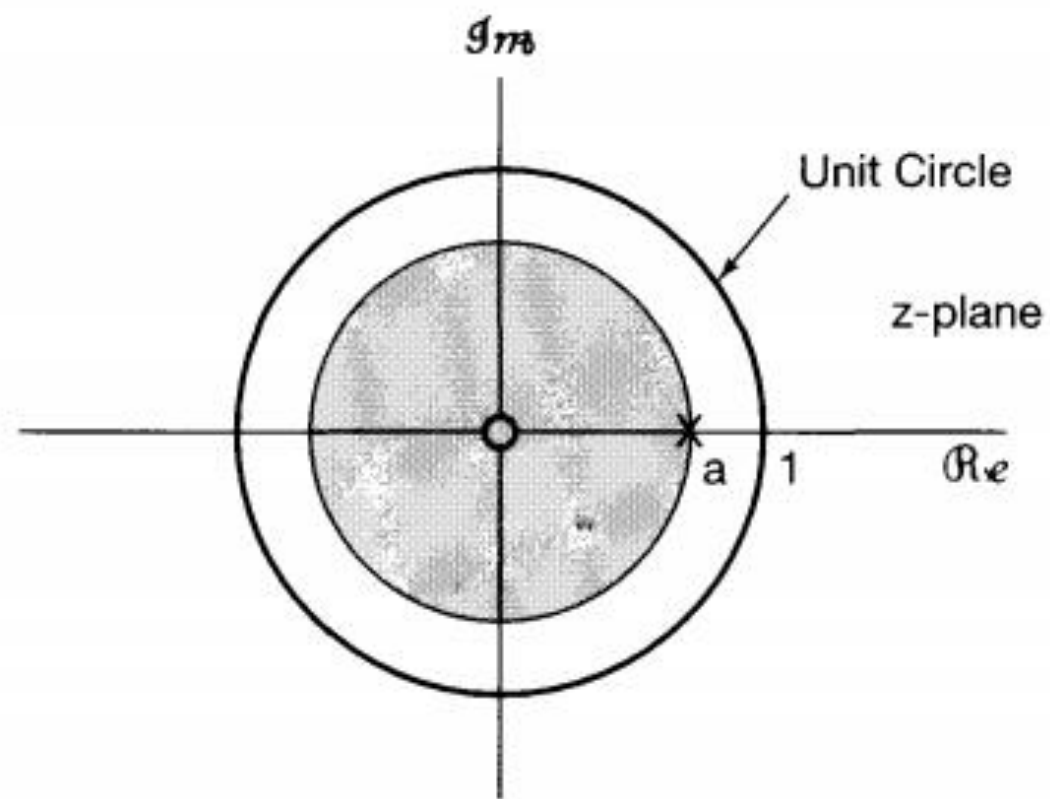


Figure 10.3 Pole-zero plot and region of convergence for Example 10.2 for $0 < a < 1$.

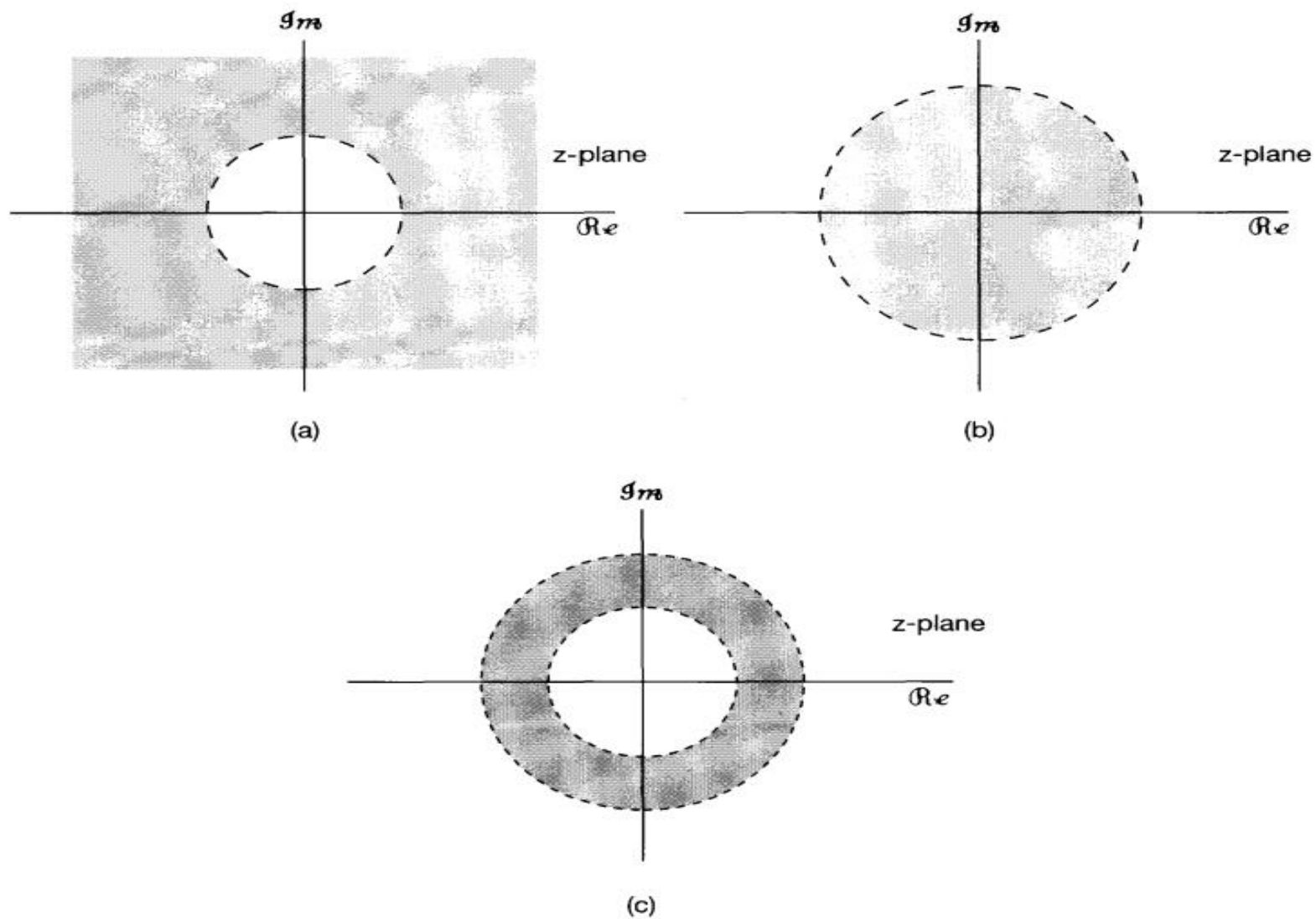


Figure 10.8 (a) ROC for right-sided sequence; (b) ROC for left-sided sequence; (c) intersection of the ROCs in (a) and (b), representing the ROC for a two-sided sequence that is the sum of the right-sided and the left-sided sequence.

A discrete-time LTI system is causal if and only if the ROC of its system function is the exterior of a circle, including infinity.

Example 10.7

Let

$$x[n] = b^{|n|}, \quad b > 0. \quad (10.31)$$

This two-sided sequence is illustrated in Figure 10.10, for both $b < 1$ and $b > 1$. The z -transform for the sequence can be obtained by expressing it as the sum of a right-sided and a left-sided sequence. We have

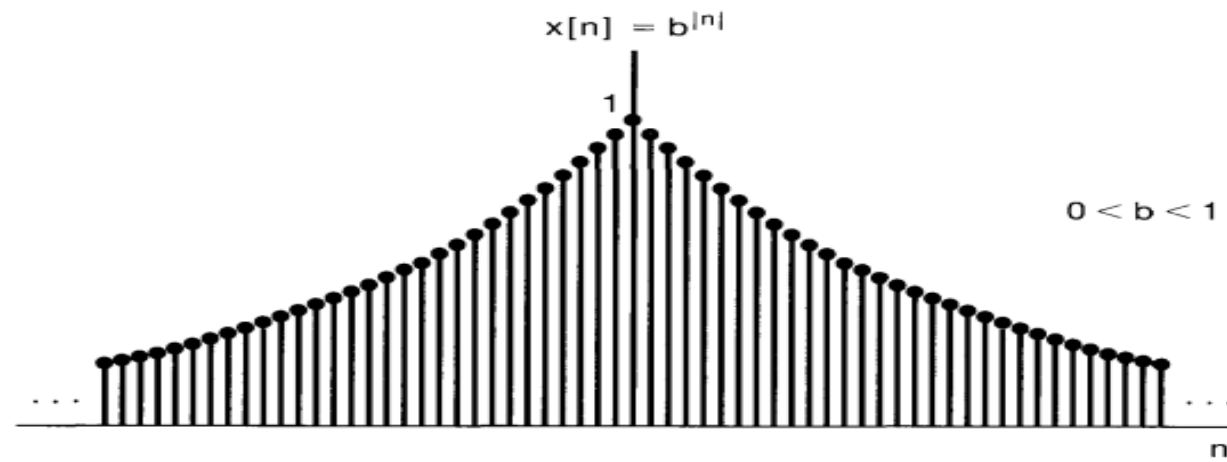
$$x[n] = b^n u[n] + b^{-n} u[-n - 1]. \quad (10.32)$$

From Example 10.1,

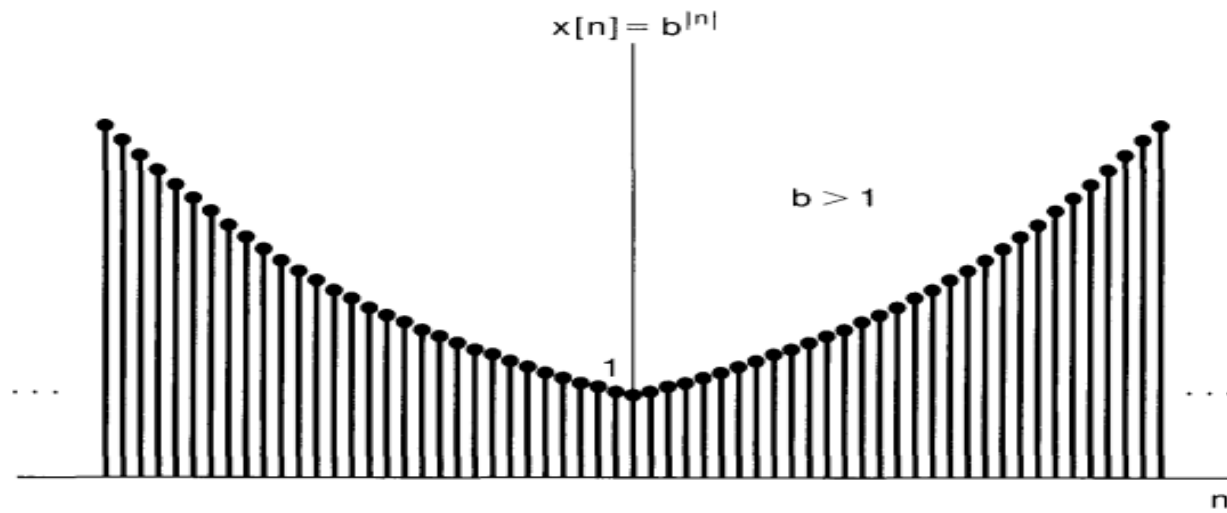
$$b^n u[n] \xleftrightarrow{z} \frac{1}{1 - bz^{-1}}, \quad |z| > b, \quad (10.33)$$

and from Example 10.2,

$$b^{-n} u[-n - 1] \xleftrightarrow{z} \frac{-1}{1 - b^{-1}z^{-1}}, \quad |z| < \frac{1}{b}. \quad (10.34)$$



(a)



(b)

Figure 10.10 Sequence $x[n] = b^{|n|}$ for $0 < b < 1$ and for $b > 1$: (a) $b = 0.95$; (b) $b = 1.05$.

$$X(z) = \frac{1}{1 - bz^{-1}} - \frac{1}{1 - b^{-1}z^{-1}}, \quad b < |z| < \frac{1}{b}, \quad (10.35)$$

or equivalently,

$$X(z) = \frac{b^2 - 1}{b} \frac{z}{(z - b)(z - b^{-1})}, \quad b < |z| < \frac{1}{b}. \quad (10.36)$$

The corresponding pole-zero pattern and ROC are shown in Figure 10.11(e).

In Figures 10.11(a)–(d) we show the pole-zero pattern and ROC for eqs. (10.33) and (10.34), for values of $b > 1$ and $0 < b < 1$. For $b > 1$, there is no common ROC, and thus the sequence in eq. (10.31) will not have a z -transform, even though the right-sided and left-sided components do individually. For $b < 1$, the ROCs in eqs. (10.33) and (10.34) overlap, and thus the z -transform for the composite sequence is

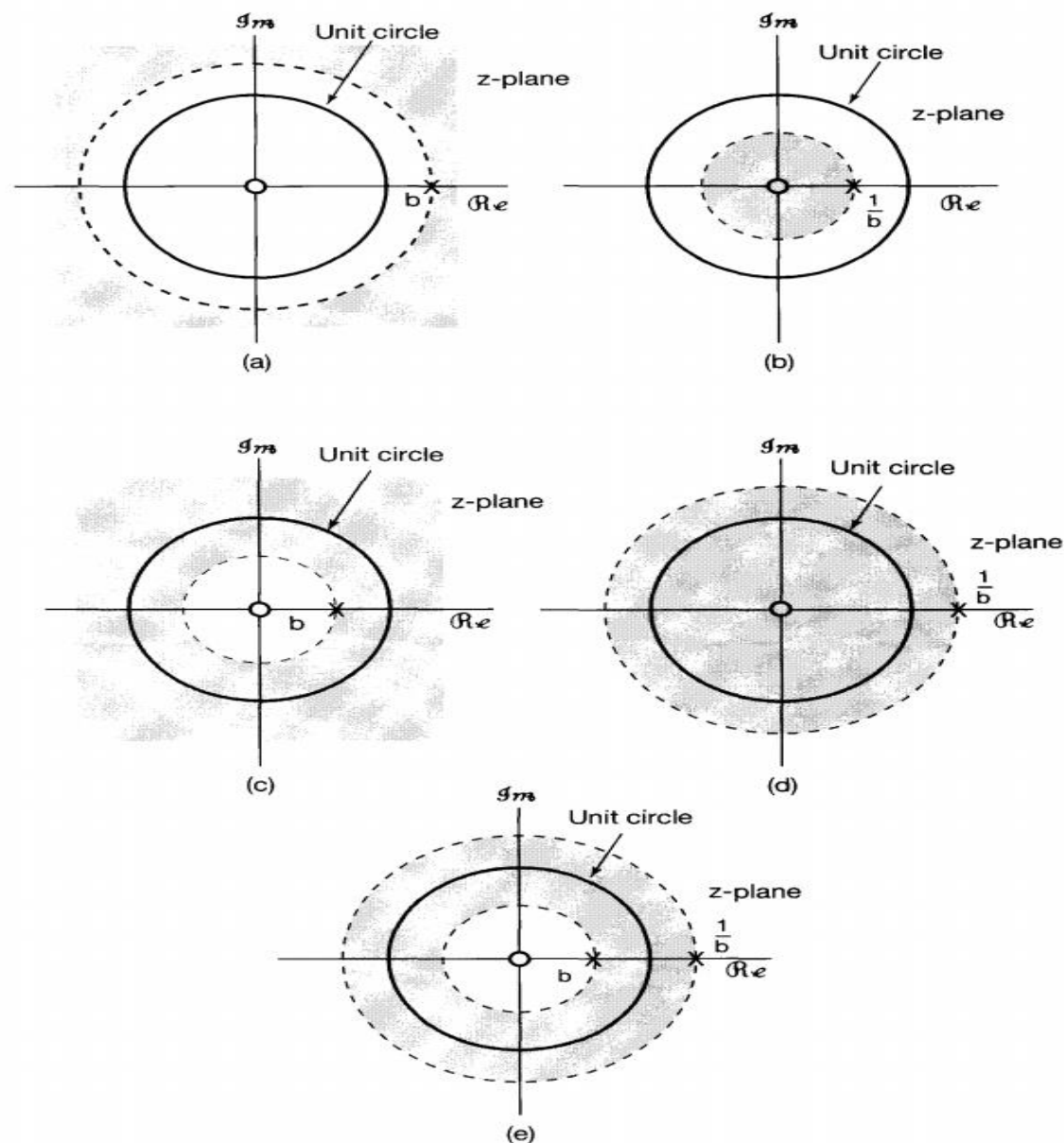


Figure 10.11 Pole-zero plots and ROCs for Example 10.7: (a) eq. (10.33) for $b > 1$; (b) eq. (10.34) for $b > 1$; (c) eq. (10.33) for $0 < b < 1$; (d) eq. (10.34) for $0 < b < 1$; (e) pole-zero plot and ROC for eq. (10.36) with $0 < b < 1$. For $b > 1$, the z-transform of $x[n]$ in eq. (10.31) does not converge for any value of z .

TABLE 10.2 SOME COMMON z -TRANSFORM PAIRS

Signal	Transform	ROC
1. $\delta[n]$	1	All z
2. $u[n]$	$\frac{1}{1 - z^{-1}}$	$ z > 1$
3. $-u[-n - 1]$	$\frac{1}{1 - z^{-1}}$	$ z < 1$
4. $\delta[n - m]$	z^{-m}	All z , except 0 (if $m > 0$) or ∞ (if $m < 0$)
5. $\alpha^n u[n]$	$\frac{1}{1 - \alpha z^{-1}}$	$ z > \alpha $
6. $-\alpha^n u[-n - 1]$	$\frac{1}{1 - \alpha z^{-1}}$	$ z < \alpha $
7. $n\alpha^n u[n]$	$\frac{\alpha z^{-1}}{(1 - \alpha z^{-1})^2}$	$ z > \alpha $
8. $-n\alpha^n u[-n - 1]$	$\frac{\alpha z^{-1}}{(1 - \alpha z^{-1})^2}$	$ z < \alpha $
9. $[\cos \omega_0 n]u[n]$	$\frac{1 - [\cos \omega_0]z^{-1}}{1 - [2 \cos \omega_0]z^{-1} + z^{-2}}$	$ z > 1$
10. $[\sin \omega_0 n]u[n]$	$\frac{[\sin \omega_0]z^{-1}}{1 - [2 \cos \omega_0]z^{-1} + z^{-2}}$	$ z > 1$
11. $[r^n \cos \omega_0 n]u[n]$	$\frac{1 - [r \cos \omega_0]z^{-1}}{1 - [2r \cos \omega_0]z^{-1} + r^2 z^{-2}}$	$ z > r$
12. $[r^n \sin \omega_0 n]u[n]$	$\frac{[r \sin \omega_0]z^{-1}}{1 - [2r \cos \omega_0]z^{-1} + r^2 z^{-2}}$	$ z > r$

TABLE 10.1 PROPERTIES OF THE z -TRANSFORM

Section	Property	Signal	z -Transform	ROC
		$x[n]$	$X(z)$	R
		$x_1[n]$	$X_1(z)$	R_1
		$x_2[n]$	$X_2(z)$	R_2
10.5.1	Linearity	$ax_1[n] + bx_2[n]$	$aX_1(z) + bX_2(z)$	At least the intersection of R_1 and R_2
10.5.2	Time shifting	$x[n - n_0]$	$z^{-n_0}X(z)$	R , except for the possible addition or deletion of the origin
10.5.3	Scaling in the z -domain	$e^{j\omega_0 n}x[n]$	$X(e^{-j\omega_0}z)$	R
		$z_0^n x[n]$	$X\left(\frac{z}{z_0}\right)$	$z_0 R$
		$a^n x[n]$	$X(a^{-1}z)$	Scaled version of R (i.e., $ a R$ = the set of points $\{ a z\}$ for z in R)
10.5.4	Time reversal	$x[-n]$	$X(z^{-1})$	Inverted R (i.e., R^{-1} = the set of points z^{-1} , where z is in R)
10.5.5	Time expansion	$x_{(k)}[n] = \begin{cases} x[r], & n = rk \\ 0, & n \neq rk \end{cases}$ for some integer r	$X(z^k)$	$R^{1/k}$ (i.e., the set of points $z^{1/k}$, where z is in R)
10.5.6	Conjugation	$x^*[n]$	$X^*(z^*)$	R
10.5.7	Convolution	$x_1[n] * x_2[n]$	$X_1(z)X_2(z)$	At least the intersection of R_1 and R_2
10.5.7	First difference	$x[n] - x[n - 1]$	$(1 - z^{-1})X(z)$	At least the intersection of R and $ z > 0$
10.5.7	Accumulation	$\sum_{k=-\infty}^n x[k]$	$\frac{1}{1 - z^{-1}}X(z)$	At least the intersection of R and $ z > 1$
10.5.8	Differentiation in the z -domain	$nx[n]$	$-z \frac{dX(z)}{dz}$	R
10.5.9	Initial Value Theorem If $x[n] = 0$ for $n < 0$, then			

$$X(z) = \frac{1}{(1 - \frac{1}{3}z^{-1})(1 - 2z^{-1})}, \quad (10.37)$$

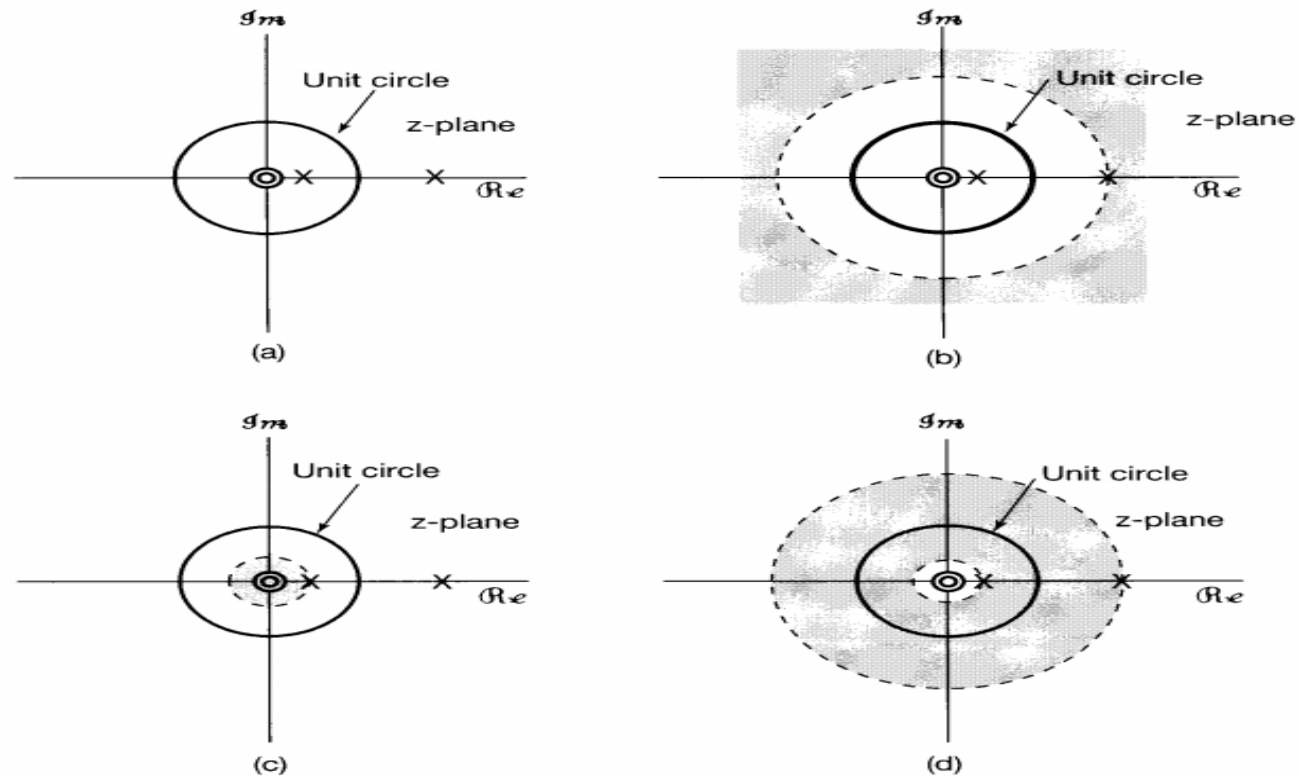


Figure 10.12 The three possible ROCs that can be connected with the expression for the z-transform in Example 10.8: (a) pole-zero pattern for $X(z)$; (b) pole-zero pattern and ROC if $x[n]$ is right sided; (c) pole-zero pattern and ROC if $x[n]$ is left sided; (d) pole-zero pattern and ROC if $x[n]$ is two sided. In each case, the zero at the origin is a second-order zero.

Inverse Z Transform

- 1) By using partial fraction method

Consider the z -transform

$$X(z) = \frac{3 - \frac{5}{6}z^{-1}}{(1 - \frac{1}{4}z^{-1})(1 - \frac{1}{3}z^{-1})}, \quad |z| > \frac{1}{3}. \quad (10.42)$$

There are two poles, one at $z = 1/3$ and one at $z = 1/4$, and the ROC lies outside the outermost pole. That is, the ROC consists of all points with magnitude greater than that of the pole with the larger magnitude, namely the pole at $z = 1/3$. From Property 4 in Section 10.2, we then know that the inverse transform is a right-sided sequence. As described in the appendix, $X(z)$ can be expanded by the method of partial fractions. For

this example, the partial-fraction expansion, expressed in polynomials in z^{-1} , is

$$X(z) = \frac{1}{1 - \frac{1}{4}z^{-1}} + \frac{2}{1 - \frac{1}{3}z^{-1}}. \quad (10.43)$$

Thus, $x[n]$ is the sum of two terms, one with z -transform $1/[1 - (1/4)z^{-1}]$ and the other with z -transform $2/[1 - (1/3)z^{-1}]$. In order to determine the inverse z -transform of each of these individual terms, we must specify the ROC associated with each. Since the ROC for $X(z)$ is outside the outermost pole, the ROC for each individual term in eq. (10.43) must also be outside the pole associated with that term. That is, the ROC for each term consists of all points with magnitude greater than the magnitude of the corresponding pole. Thus,

$$x[n] = x_1[n] + x_2[n], \quad (10.44)$$

where

$$x_1[n] \xleftrightarrow{z} \frac{1}{1 - \frac{1}{4}z^{-1}}, \quad |z| > \frac{1}{4}, \quad (10.45)$$

$$x_2[n] \xleftrightarrow{z} \frac{2}{1 - \frac{1}{3}z^{-1}}, \quad |z| > \frac{1}{3}. \quad (10.46)$$

From Example 10.1, we can identify by inspection that

$$x_1[n] = \left(\frac{1}{4}\right)^n u[n] \quad (10.47)$$

and

$$x_2[n] = 2\left(\frac{1}{3}\right)^n u[n], \quad (10.48)$$

and thus,

$$x[n] = \left(\frac{1}{4}\right)^n u[n] + 2\left(\frac{1}{3}\right)^n u[n]. \quad (10.49)$$

Power series expansion

Consider the z -transform

$$X(z) = 4z^2 + 2 + 3z^{-1}, \quad 0 < |z| < \infty. \quad (10.56)$$

From the power-series definition of the z -transform in eq. (10.3), we can determine the inverse transform of $X(z)$ by inspection:

$$x[n] = \begin{cases} 4, & n = -2 \\ 2, & n = 0 \\ 3, & n = 1 \\ 0, & \text{otherwise} \end{cases}.$$

That is,

$$x[n] = 4\delta[n + 2] + 2\delta[n] + 3\delta[n - 1]. \quad (10.57)$$

Comparing eqs. (10.56) and (10.57), we see that different powers of z serve as placeholders for sequence values at different points in time; i.e., if we simply use the transform pair

$$\delta[n + n_0] \xleftrightarrow{z} z^{n_0},$$

we can immediately pass from eq. (10.56) to (10.57) and vice versa.

Example 10.13

Consider

$$X(z) = \frac{1}{1 - az^{-1}}, \quad |z| > |a|.$$

This expression can be expanded in a power series by long division:

$$\begin{array}{r} 1 + az^{-1} + a^2z^{-2} + \cdots \\ 1 - az^{-1} \overline{) 1} \\ \underline{1 - az^{-1}} \phantom{+ a^2z^{-2} + \cdots} \\ az^{-1} - a^2z^{-2} \\ \underline{az^{-1} - a^2z^{-2}} \\ a^2z^{-2} \\ \vdots \end{array},$$

or

$$\frac{1}{1 - az^{-1}} = 1 + az^{-1} + a^2z^{-2} + \dots \quad (10.58)$$

The series expansion of eq. (10.58) converges, since $|z| > |a|$, or equivalently, $|az^{-1}| < 1$. Comparing this equation with the definition of the z -transform in equation (10.3), we see, by matching terms in powers of z , that $x[n] = 0, n < 0$; $x[0] = 1$; $x[1] = a$; $x[2] = a^2$; and in general, $x[n] = a^n u[n]$, which is consistent with Example 10.1.

If, instead, the ROC of $X(z)$ is specified as $|z| < |a|$ or, equivalently, $|az^{-1}| > 1$, then the power-series expansion for $1/(1 - az^{-1})$ in eq. (10.58) does not converge. However, we can obtain a convergent power series by long division again:

$$\begin{array}{r} -a^{-1}z - a^{-2}z^2 - \dots \\ -az^{-1} + 1 \overline{) 1} \\ \underline{1 - a^{-1}z} \\ a^{-1}z \end{array},$$

or

$$\frac{1}{1 - az^{-1}} = -a^{-1}z - a^{-2}z^2 - \dots \quad (10.59)$$

In this case, then, $x[n] = 0$, $n \geq 0$; and $x[-1] = -a^{-1}$, $x[-2] = -a^{-2}$, $\dots$; that is, $x[n] = -a^n u[-n - 1]$. This is consistent with Example 10.2.

The power-series expansion method for obtaining the inverse z -transform is particularly useful for nonrational z -transforms, which we illustrate with the following example:

Example 10.14

Consider the z -transform

$$X(z) = \log(1 + az^{-1}), \quad |z| > |a|. \quad (10.60)$$

With $|z| > |a|$, or, equivalently, $|az^{-1}| < 1$, eq. (10.60) can be expanded in a power series using the Taylor's series expansion

$$\log(1 + v) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1} v^n}{n}, \quad |v| < 1. \quad (10.61)$$

Applying this to eq. (10.60), we have

$$X(z) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1} a^n z^{-n}}{n}, \quad (10.62)$$

from which we can identify

$$x[n] = \begin{cases} (-1)^{n+1} \frac{a^n}{n}, & n \geq 1, \\ 0, & n \leq 0 \end{cases}, \quad (10.63)$$

or equivalently,

$$x[n] = \frac{-(-a)^n}{n} u[n-1].$$

In Problem 10.63 we consider a related example with region of convergence $|z| < |a|$.

10.5.9 The Initial-Value Theorem

If $x[n] = 0, n < 0$, then

$$x[0] = \lim_{z \rightarrow \infty} X(z). \quad (10.95)$$

This property follows by considering the limit of each term individually in the expression for the z -transform, with $x[n]$ zero for $n < 0$. With this constraint,

$$X(z) = \sum_{n=0}^{\infty} x[n]z^{-n}.$$

As $z \rightarrow \infty$, $z^{-n} \rightarrow 0$ for $n > 0$, whereas for $n = 0$, $z^{-n} = 1$. Thus, eq. (10.95) follows.

As one consequence of the initial-value theorem, for a causal sequence, if $x[0]$ is finite, then $\lim_{z \rightarrow \infty} X(z)$ is finite. Consequently, with $X(z)$ expressed as a ratio of polynomials in z , the order of the numerator polynomial cannot be greater than the order of the denominator polynomial; or, equivalently, the number of finite zeros of $X(z)$ cannot be greater than the number of finite poles.

Initial Value and Final Value Theorems

- Initial value and final value theorems of z-transform are defined for causal signal.
- Initial Value Theorem: $x(0) = \lim_{z \rightarrow \infty} X(z)$
- Final Value Theorem $x(\infty) = \lim_{z \rightarrow 1} [z-1]X(z)$

Causal System

A discrete-time LTI system with rational system function $H(z)$ is causal if and only if: (a) the ROC is the exterior of a circle outside the outermost pole; and (b) with $H(z)$ expressed as a ratio of polynomials in z , the order of the numerator cannot be greater than the order of the denominator.

Example 10.20

Consider a system with system function whose algebraic expression is

$$H(z) = \frac{z^3 - 2z^2 + z}{z^2 + \frac{1}{4}z + \frac{1}{8}}.$$

Without even knowing the ROC for this system, we can conclude that the system is not causal, because the numerator of $H(z)$ is of higher order than the denominator.

Example 10.21

Consider a system with system function

$$H(z) = \frac{1}{1 - \frac{1}{2}z^{-1}} + \frac{1}{1 - 2z^{-1}}, \quad |z| > 2 \quad (10.97)$$

Since the ROC for this system function is the exterior of a circle outside the outermost pole, we know that the impulse response is right-sided. To determine if the system is causal, we then need only check the other condition required for causality, namely that $H(z)$, when expressed as a ratio of polynomials in z , has numerator degree no larger than the denominator. For this example,

$$H(z) = \frac{2 - \frac{5}{2}z^{-1}}{(1 - \frac{1}{2}z^{-1})(1 - 2z^{-1})} = \frac{2z^2 - \frac{5}{2}z}{z^2 - \frac{5}{2}z + 1}, \quad (10.98)$$

so that the numerator and denominator of $H(z)$ are both of degree two, and consequently we can conclude that the system is causal. This can also be verified by calculating the inverse transform of $H(z)$. In particular, using transform pair 5 in Table 10.2, we find that the impulse response of this system is

$$h[n] = \left[\left(\frac{1}{2} \right)^n + 2^n \right] u[n]. \quad (10.99)$$

Since $h[n] = 0$ for $n < 0$, we can confirm that the system is causal.

Stable System

An LTI system is stable if and only if the ROC of its system function $H(z)$ includes the unit circle, $|z| = 1$.

A causal LTI system with rational system function $H(z)$ is stable if and only if all of the poles of $H(z)$ lie inside the unit circle—i.e., they must all have magnitude smaller than 1.

Example 10.23

Consider a causal system with system function

$$H(z) = \frac{1}{1 - az^{-1}},$$

which has a pole at $z = a$. For this system to be stable, its pole must be inside the unit circle, i.e., we must have $|a| < 1$. This is consistent with the condition for the absolute summability of the corresponding impulse response $h[n] = a^n u[n]$.

LTI Systems Characterized by linear Constant-Coefficient Difference Equations

Example 10.25

Consider an LTI system for which the input $x[n]$ and output $y[n]$ satisfy the linear constant-coefficient difference equation

$$y[n] - \frac{1}{2}y[n-1] = x[n] + \frac{1}{3}x[n-1]. \quad (10.102)$$

Applying the z -transform to both sides of eq. (10.102), and using the linearity property set forth in Section 10.5.1 and the time-shifting property presented in Section 10.5.2, we obtain

$$Y(z) - \frac{1}{2}z^{-1}Y(z) = X(z) + \frac{1}{3}z^{-1}X(z),$$

or

$$Y(z) = X(z) \left[\frac{1 + \frac{1}{3}z^{-1}}{1 - \frac{1}{2}z^{-1}} \right]. \quad (10.103)$$

From eq. (10.96), then,

$$H(z) = \frac{Y(z)}{X(z)} = \frac{1 + \frac{1}{3}z^{-1}}{1 - \frac{1}{2}z^{-1}}. \quad (10.104)$$

This provides the algebraic expression for $H(z)$, but not the region of convergence. In fact, there are two distinct impulse responses that are consistent with the difference equation (10.102), one right sided and the other left sided. Correspondingly, there are two different choices for the ROC associated with the algebraic expression (10.104). One, $|z| > 1/2$, is associated with the assumption that $h[n]$ is right sided, and the other, $|z| < 1/2$, is associated with the assumption that $h[n]$ is left sided.

Consider first the choice of ROC equal to $|z| > 1$. Writing

$$H(z) = \left(1 + \frac{1}{3}z^{-1}\right) \frac{1}{1 - \frac{1}{2}z^{-1}},$$

we can use transform pair 5 in Table 10.2, together with the linearity and time-shift properties, to find the corresponding impulse response

$$h[n] = \left(\frac{1}{2}\right)^n u[n] + \frac{1}{3} \left(\frac{1}{2}\right)^{n-1} u[n-1].$$

For the other choice of ROC, namely, $|z| < 1$, we can use transform pair 6 Table 10.2 and the linearity and time-shifting properties, yielding

$$h[n] = -\left(\frac{1}{2}\right)^n u[-n-1] - \frac{1}{3} \left(\frac{1}{2}\right)^{n-1} u[-n].$$

In this case, the system is anticausal ($h[n] = 0$ for $n > 0$) and unstable.

Example 10.26

Suppose that we are given the following information about an LTI system:

1. If the input to the system is $x_1[n] = (1/6)^n u[n]$, then the output is

$$y_1[n] = \left[a \left(\frac{1}{2} \right)^n + 10 \left(\frac{1}{3} \right)^n \right] u[n],$$

where a is a real number.

2. If $x_2[n] = (-1)^n$, then the output is $y_2[n] = \frac{7}{4}(-1)^n$. As we now show, from these two pieces of information, we can determine the system function $H(z)$ for this system, including the value of the number a , and can also immediately deduce a number of other properties of the system.

The z -transforms of the signals specified in the first piece of information are

$$X_1(z) = \frac{1}{1 - \frac{1}{6}z^{-1}}, \quad |z| > \frac{1}{6}, \quad (10.107)$$

$$\begin{aligned} Y_1(z) &= \frac{a}{1 - \frac{1}{2}z^{-1}} + \frac{10}{1 - \frac{1}{3}z^{-1}} \\ &= \frac{(a + 10) - (5 + \frac{a}{3})z^{-1}}{(1 - \frac{1}{2}z^{-1})(1 - \frac{1}{3}z^{-1})}, \quad |z| > \frac{1}{2}. \end{aligned} \quad (10.108)$$

From eq. (10.96), it follows that the algebraic expression for the system function is

$$H(z) = \frac{Y_1(z)}{X_1(z)} = \frac{[(a + 10) - (5 + \frac{a}{3})z^{-1}][1 - \frac{1}{6}z^{-1}]}{(1 - \frac{1}{2}z^{-1})(1 - \frac{1}{3}z^{-1})}. \quad (10.109)$$

Furthermore, we know that the response to $x_2[n] = (-1)^n$ must equal $(-1)^n$ multiplied by the system function $H(z)$ evaluated at $z = -1$. Thus from the second piece of information given, we see that

$$\frac{7}{4} = H(-1) = \frac{[(a + 10) + 5 + \frac{a}{3}][\frac{7}{6}]}{(\frac{3}{2})(\frac{4}{3})}. \quad (10.110)$$

Solving eq. (10.110), we find that $a = -9$, so that

$$H(z) = \frac{(1 - 2z^{-1})(1 - \frac{1}{6}z^{-1})}{(1 - \frac{1}{2}z^{-1})(1 - \frac{1}{3}z^{-1})}, \quad (10.111)$$

or

$$H(z) = \frac{1 - \frac{13}{6}z^{-1} + \frac{1}{3}z^{-2}}{1 - \frac{5}{6}z^{-1} + \frac{1}{6}z^{-2}}, \quad (10.112)$$

or, finally,

$$H(z) = \frac{z^2 - \frac{13}{6}z + \frac{1}{3}}{z^2 - \frac{5}{6}z + \frac{1}{6}}. \quad (10.113)$$

Also, from the convolution property, we know that the ROC of $Y_1(z)$ must include at least the intersections of the ROCs of $X_1(z)$ and $H(z)$. Examining the three possible ROCs for $H(z)$ (namely, $|z| < 1/3$, $1/3 < |z| < 1/2$, and $|z| > 1/2$), we find that the only choice that is consistent with the ROCs of $X_1(z)$ and $Y_1(z)$ is $|z| > 1/2$.

Since the ROC for the system includes the unit circle, we know that the system is stable. Furthermore, from eq. (10.113) with $H(z)$ viewed as a ratio of polynomials in z , the order of the numerator does not exceed that of the denominator, and thus we can conclude that the LTI system is causal. Also, using eqs. (10.112) and (10.106), we can write the difference equation that, together with the condition of initial rest, characterizes the system:

$$y[n] - \frac{5}{6}y[n-1] + \frac{1}{6}y[n-2] = x[n] - \frac{13}{6}x[n-1] + \frac{1}{3}x[n-2].$$